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Artificial Intelligence in Genomic Medicine

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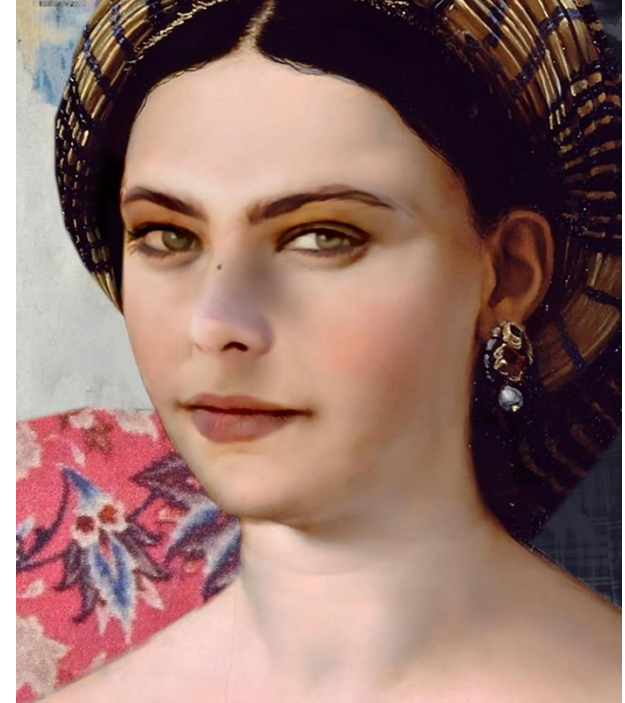
What is Artificial Intelligence?

- Artificial Intelligence (AI) is the theory and development of computer systems able to perform tasks that normally require human intelligence, such as visual perception, speech recognition, decision-making, and other complex tasks



Examples of advanced AI appearing throughout society

- Manufacturing robots
- Automated financial investing
- Social media monitoring
- Smart assistants
- Conversational bots
 - ChatGPT



Many others...



Applications in biomedical field far too many to list

- Medical image analysis
 - MRI segmentation
 - Histopathological cancer classification
 - Mammographic lesion detection
- Protein structure prediction
- Genome analysis
 - Gene expression inference
 - Enhancer prediction
 - Variant pathogenicity prediction

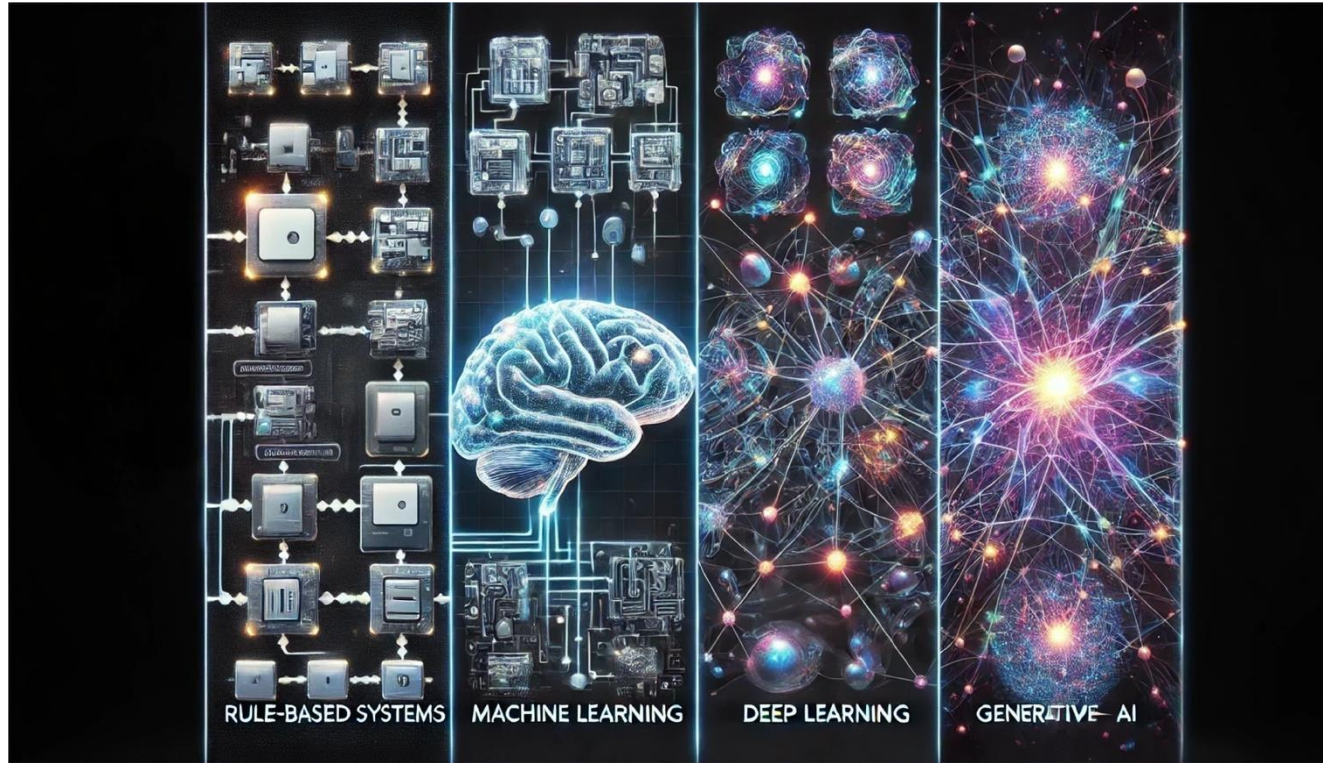


What are the key challenges of AI?

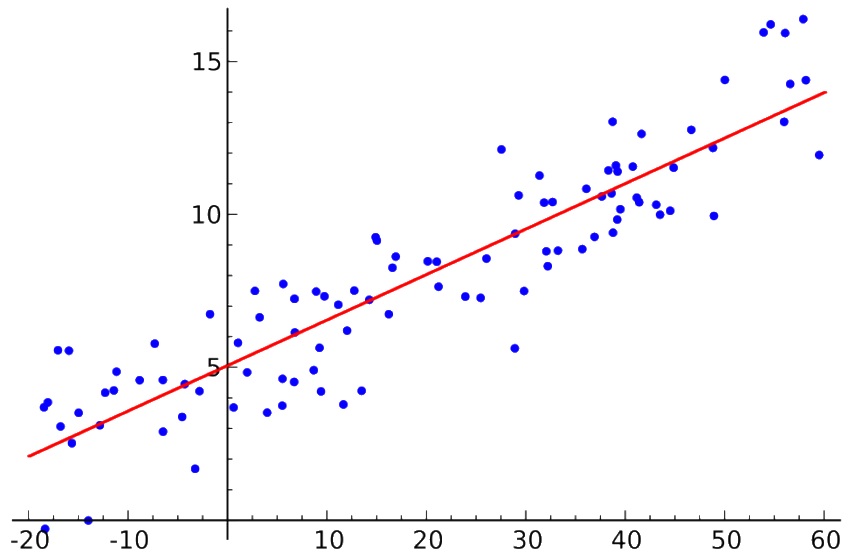
- Key Challenges of AI: knowledge representation, natural language processing (NLP), learning, perception, planning, reasoning, motion, social intelligence and general intelligence
 - Knowledge representation - catalog concepts and relationships using knowledgebases, ontologies, and structured data, for computational use
 - Natural language processing (e.g., text-mining) - allow computers to read and understand human language
 - Learning (e.g., machine learning) - develop computer algorithms that improve automatically through experience
 - Unsupervised - Find novel patterns in data without human input or labels
 - Supervised - Predict unknowns from labeled knowns (e.g., classifiers)



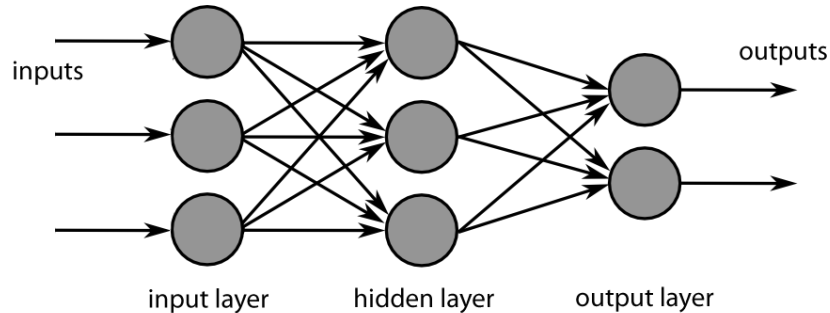
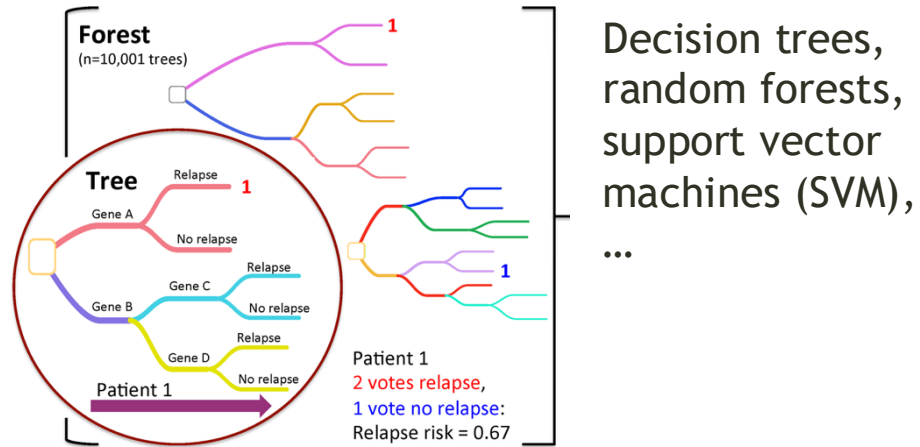
AI has gone through several eras



Learning -> Machine Learning -> Deep Learning



Linear regression



multi-layered artificial neural networks



General considerations

- Establish training and testing datasets
 - Are there batch effects to consider?
 - Randomly split data
 - Make sure balanced for key features (sex, target class, etc)
 - Identify independent test sets
- Define target classes (e.g., Relapse vs No Relapse)
 - How is relapse defined?
 - Are the classes equal sizes? If not, how to mitigate?
- Are there features which should be filtered out?
- Are there new features we wish to design/engineer?
- Is the model overfit/overtrained?



Let's walk through a classic machine learning example

Griffith *et al.* *Genome Medicine* 2013, **5**:92
<http://genomemedicine.com/content/5/10/92>



RESEARCH

Open Access

A robust prognostic signature for hormone-positive node-negative breast cancer

Obi L Griffith^{1,2*}, François Pepin^{1,3}, Oana M Enache¹, Laura M Heiser^{1,4}, Eric A Collisson⁵, Paul T Spellman^{1,6*}
and Joe W Gray^{1,4*}

<https://www.biostars.org/p/85124/>



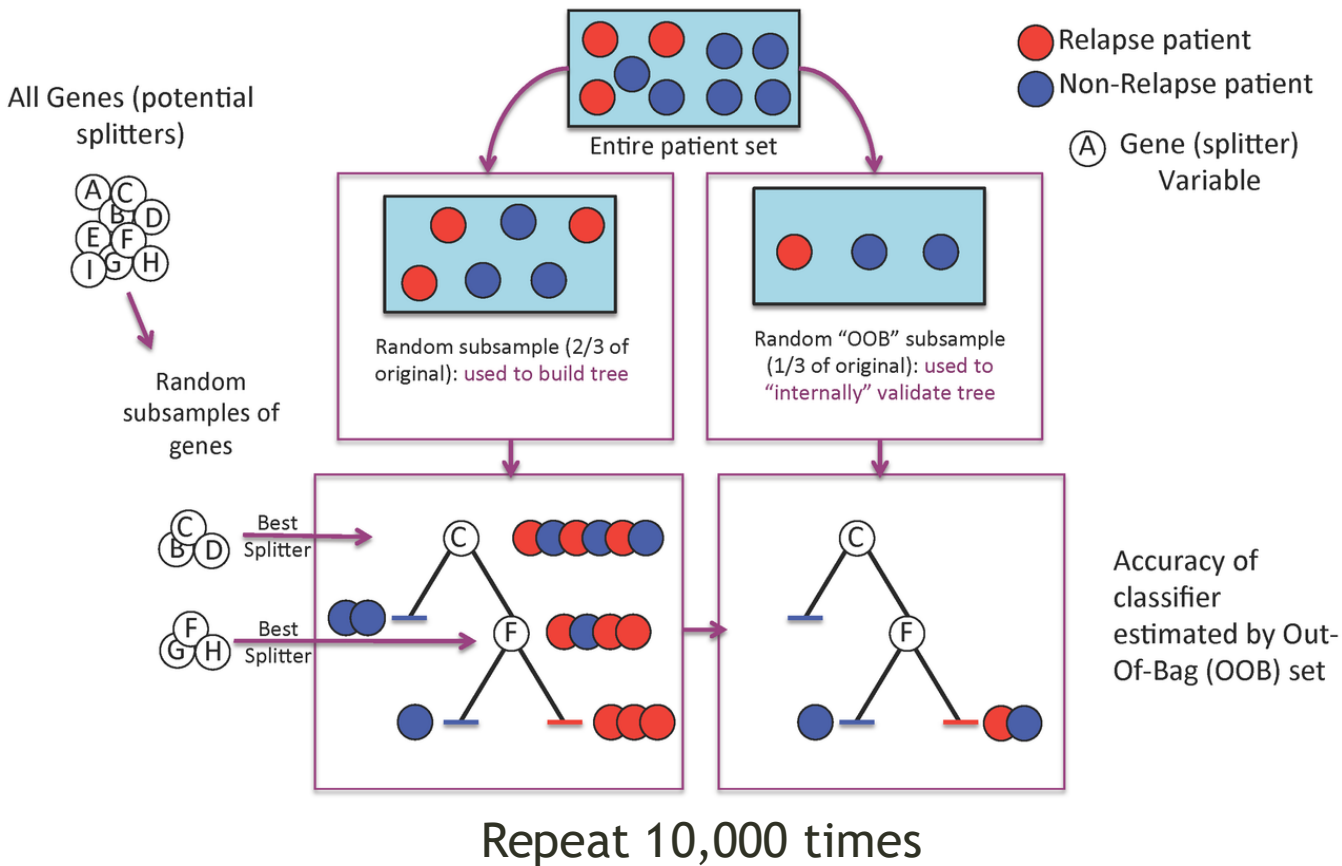
Let's walk through an example

- Let's build a classifier of relapse in ER+ breast cancer
- Suppose we have a gene expression dataset for a set of patients, with known relapse status

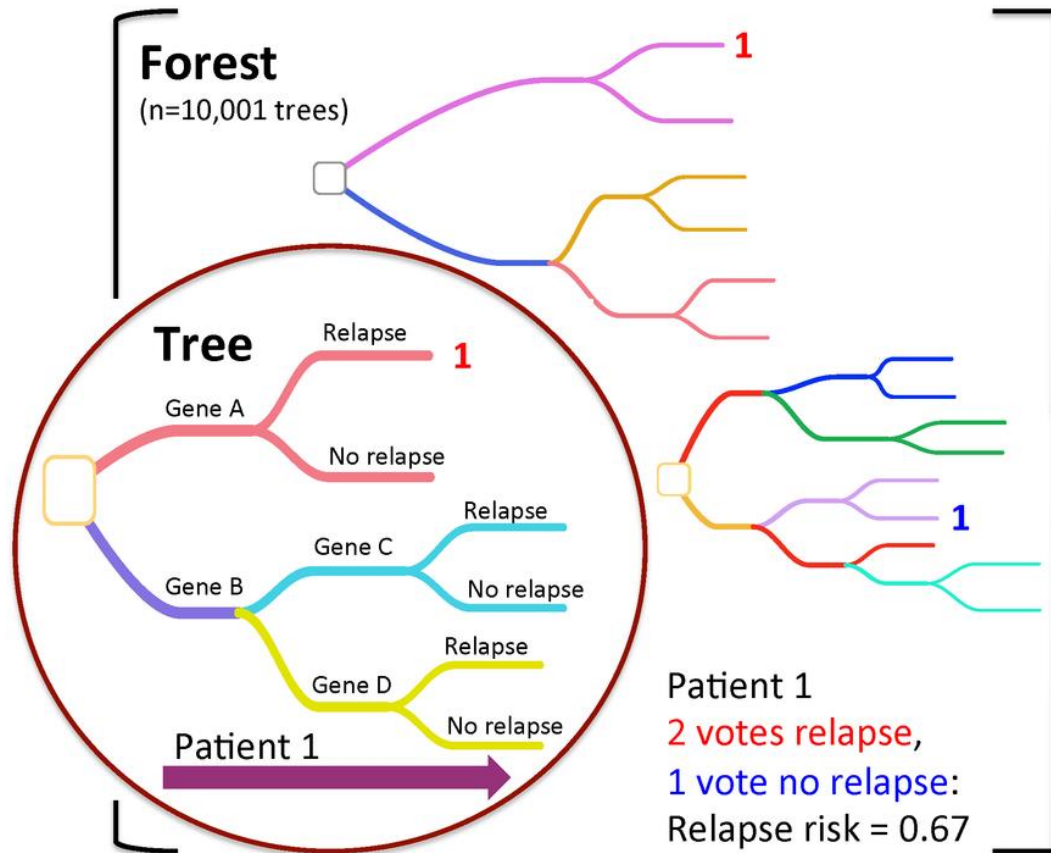
	Relapse	Relapse	No Relapse	No Relapse
	Patient1	Patient2	Patient3	Patient4
GeneA	4.7	5.2	1.1	0.9
GeneB	0.5	0.7	6.8	7.1
GeneC	2.3	3.1	2.4	3.2



How do you build (train) a Random Forest model?



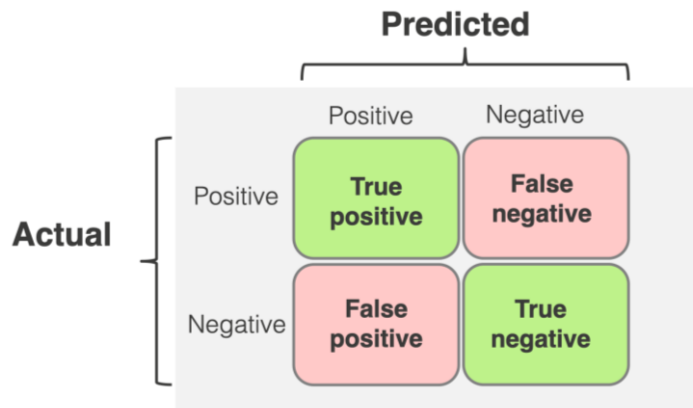
How do you predict with a Random Forest model?



- Creates a forest of many binary decision trees
 - Each Patient traverses each tree until it reaches a terminal node
 - At the terminal node each tree casts a vote (eg. "relapse"); the proportion of relapse votes from all votes is that patient's predicted relapse risk
- Class with most votes is assigned



How does our model perform? Let's look at the dreaded confusion matrix



Actual	Predicted	
	Relapse	NoRelapse
Relapse	49	58
NoRelapse	39	140

Sensitivity/Recall (true positive rate) is
how well a test can identify true positives

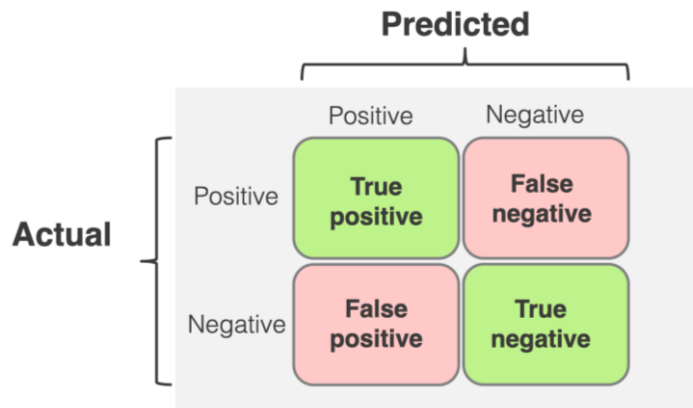
$$49 / (49 + 58) = 0.458$$

Specificity/Selectivity (true negative rate)
is how well a test can identify true negatives

$$140 / (39 + 140) = 0.782$$



How does our model perform? Let's look at the dreaded confusion matrix



Actual	Predicted	
	Relapse	NoRelapse
Relapse	49	58
NoRelapse	39	140

Relapse class error rate is how many true Relapses were miscalled

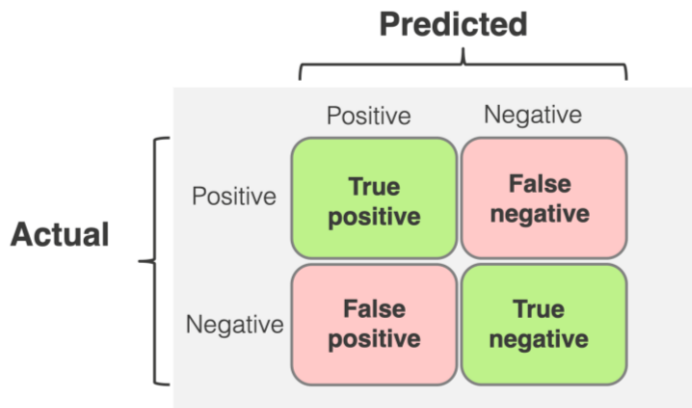
$$58 / (49 + 58) = 0.542$$

No Relapse class error rate is how many true No Relapses were miscalled

$$39 / (39 + 140) = 0.218$$



How does our model perform? Let's look at the dreaded confusion matrix



Overall accuracy

Overall error

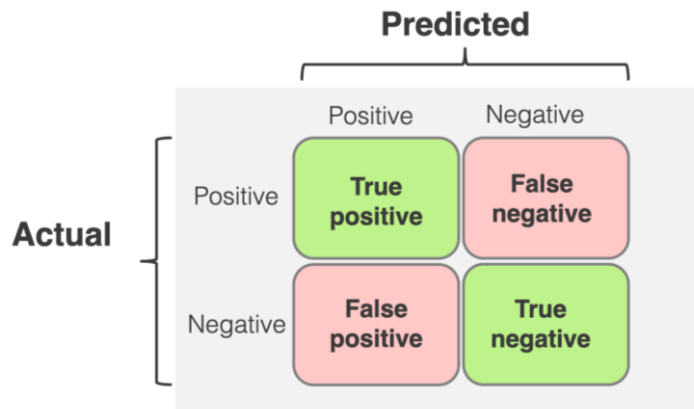
Actual	Predicted	
	Relapse	NoRelapse
Relapse	49	58
NoRelapse	39	140

$$(49+140)/(49+58+39+140) = 0.660$$

$$(39+58)/(49+58+39+140) = 0.340$$



How does our model perform? Let's look at the dreaded confusion matrix



Actual	Predicted	
	Relapse	NoRelapse
Relapse	49	58
NoRelapse	39	140

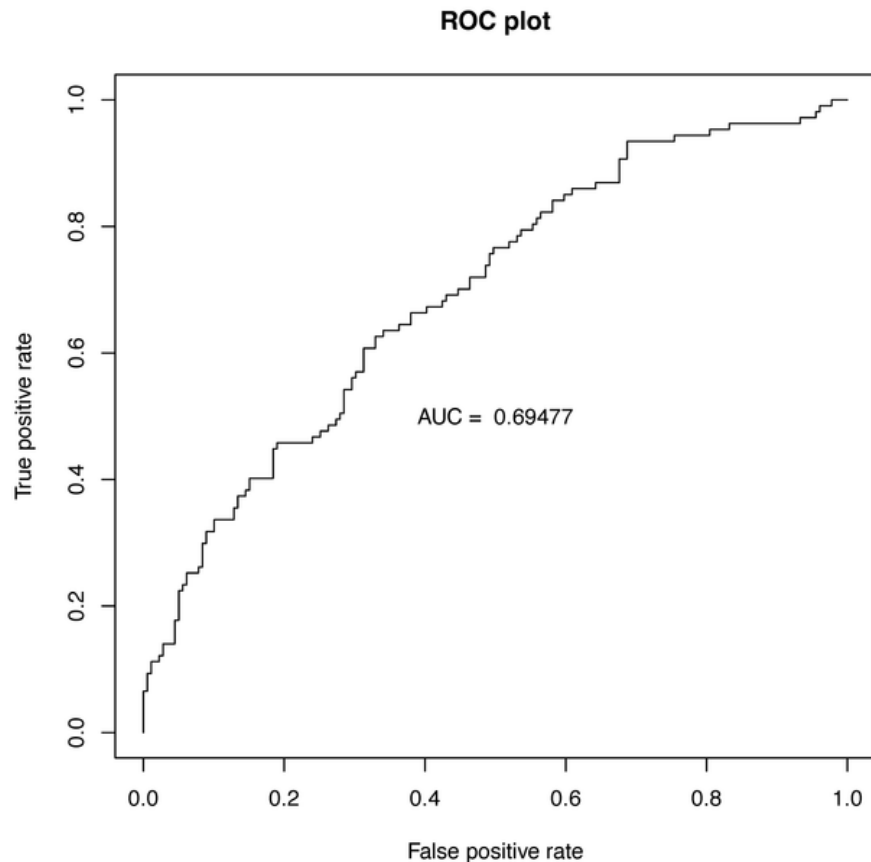
Precision is the share of true positive predictions in all positive predictions

$$49 / (49 + 39) = 0.557$$

Other popular metrics: F1 score, PPV, NPV, etc



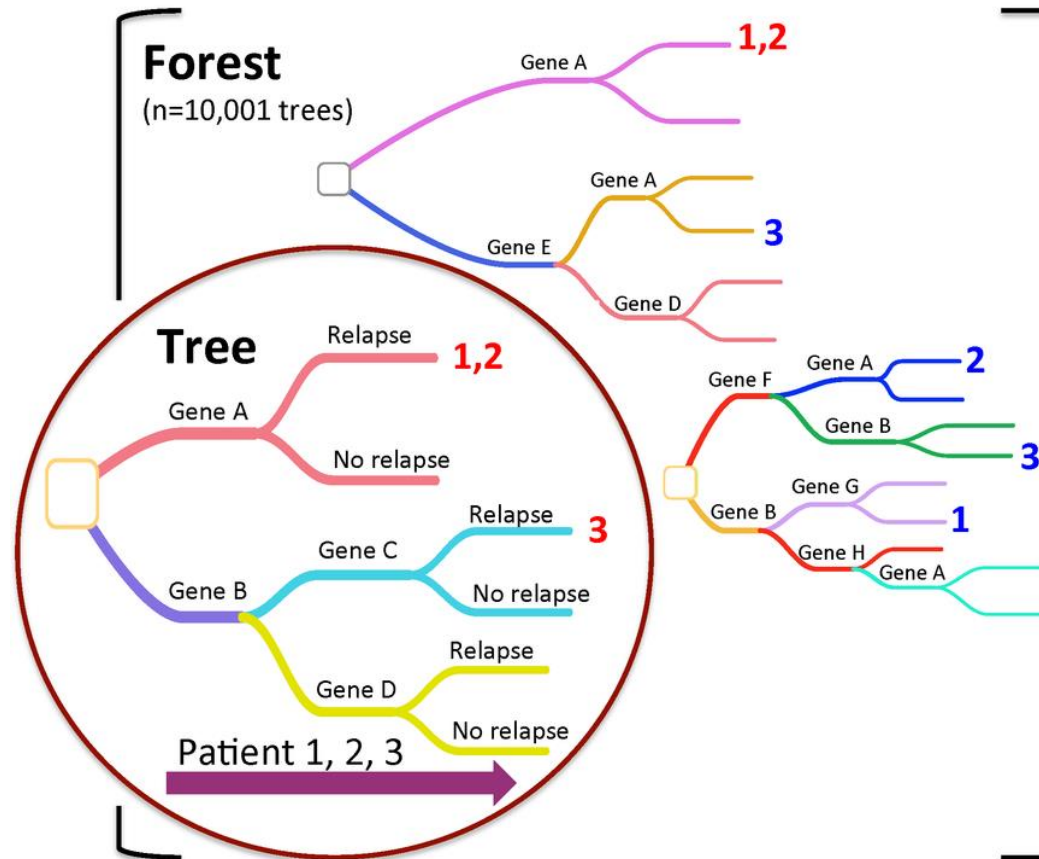
Receiver Operating Characteristic (ROC) plots TPR (Sensitivity) vs FPR for all possible model thresholds



The ROC AUC summarizes how well the model can distinguish between positive and negative classes



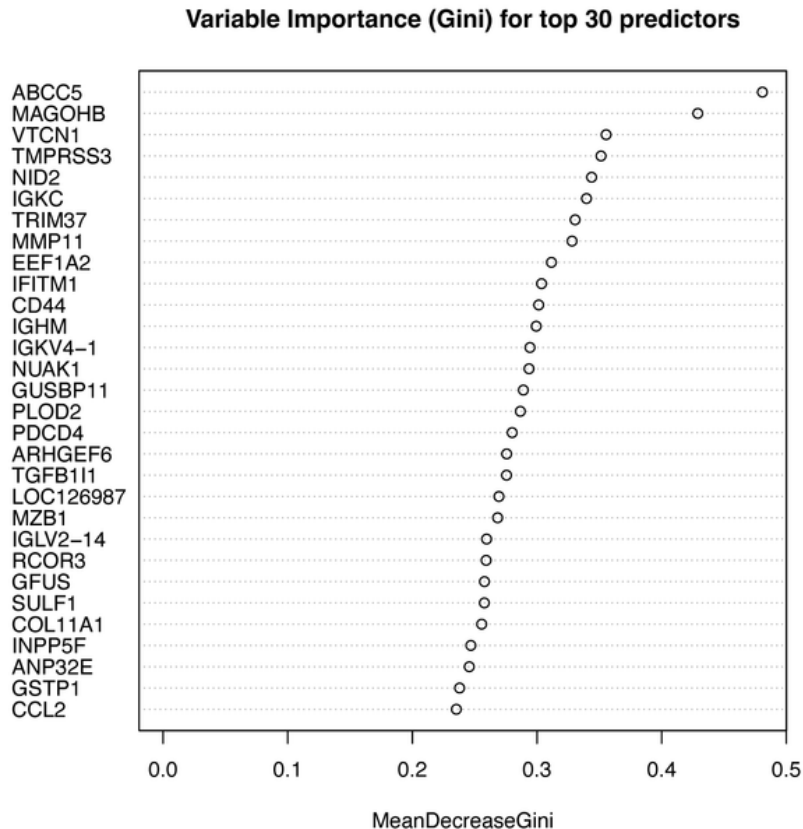
How do we know which features are important in our model



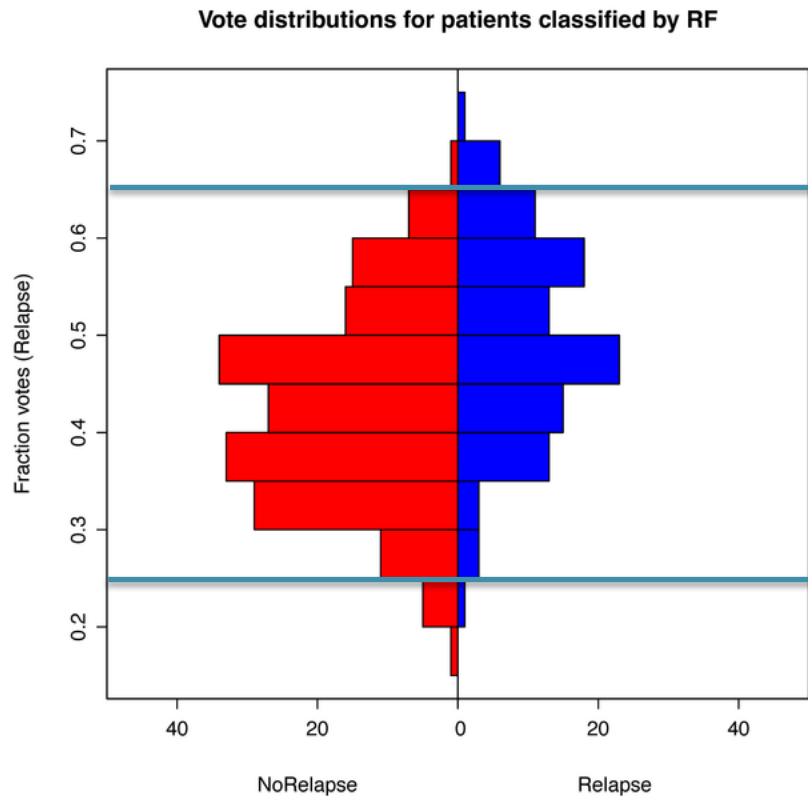
- The more often a gene is chosen as a splitter variable, the higher its “Variable Importance” – This can be used to prioritize which genes to select for an assay with limited gene measurements

Gene	Var. Imp.
Gene A	0.67
Gene B	0.20
Gene D	0.13
...	...

Variable importance could be used to prioritize genes for a targeted assay



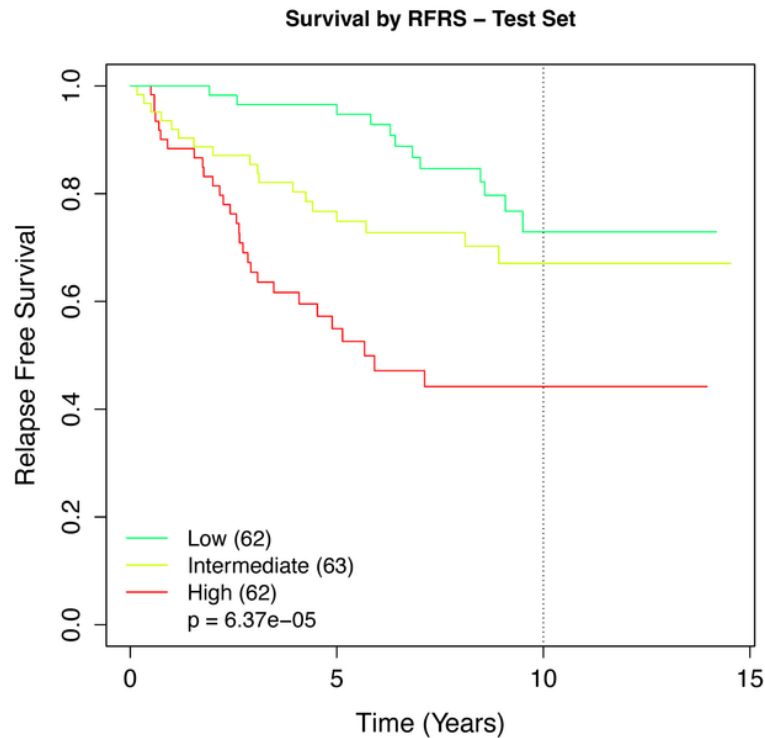
Even a model with poor overall performance may be very accurate for a subset of cases



high-risk,
intermediate-risk and
low-risk groups can be
defined



These risk groups can be related back to survival



This is how essentially how Oncotype DX, MammaPrint, and ProSigna assays were developed - used routinely in breast cancer care



Practical exercise on machine learning for cancer classification



Community

Planet

New Post

Bioinformatics Handbook Short lessons for large-scale genomics

CLICK HERE

Tutorial: Machine Learning For Cancer Classification - Part 1 - Preparing The Data Sets



This tutorial is part of a series illustrating basic concepts and techniques for machine learning in R. We will try to build a classifier of relapse in breast cancer. The analysis plan will follow the general pattern (simplified) of [a past publication](#).

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The first step is to prepare the data sets. We will use [GSE2034](#) as a training data set and [GSE2990](#) as a test data set. These data sets both make use of the Affymetrix [U133A](#) platform ([GPL96](#)). First, let's download and pre-process the training data. Note: The gcrma step may require you to have as much as ~8gb of ram. I ran this tutorial most recently on a MacBook Pro (Sonoma) with R 4.4.0 installed. It should also work on linux or windows but package installation might differ slightly.

Set some data directories.

```
tempdir="/Users/obigriffith/temp"
outdir="/Users/obigriffith/git/biostar-tutorials/MachineLearning"
```

Install the necessary bioconductor packages, if not already installed

```
if (!require("BiocManager", quietly = TRUE))
  install.packages("BiocManager")
```

```
BiocManager::install(version = "3.19")
BiocManager::install("GEOquery")
BiocManager::install("affy")
BiocManager::install("gcrma")
BiocManager::install("org.Hs.eg.db")
```



12.4 years ago

[Obi Griffith](#) ⚡ 20k



<https://www.biostars.org/p/85124/>

Nick Spies



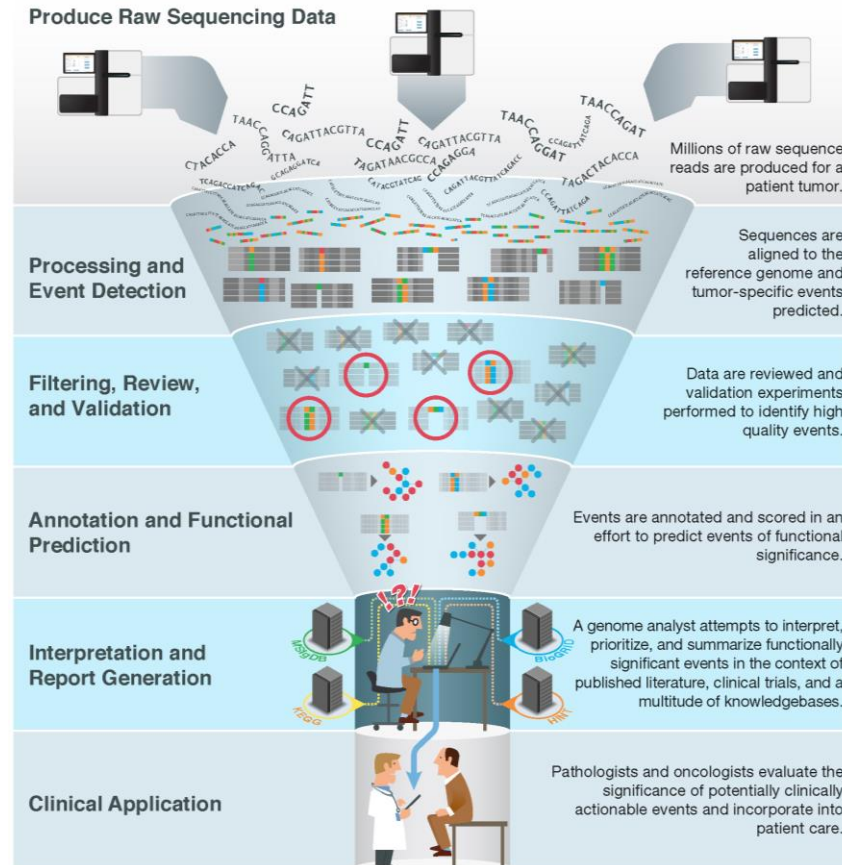
Practical exercise on machine learning for cancer classification

- It is organized into 4 parts:
- [Part 1](#) - Preparing The Data Sets
 - Download microarray expression datasets, clinical data, and normalize for training and testing
- [Part 2](#) - Building A Random Forest Classifier
 - Build a RandomForest model and assess its performance and feature importance
- [Part 3](#) - Predicting With A Random Forest Classifier
 - Apply model from Part 2 to test set, assess performance
- [Part 4](#) - Plotting A Kaplan-Meier Curve For Survival Analysis
 - Classify patients into “risk groups” and assess for survival differences

<https://www.biostars.org/p/85124/>



How else can AI be applied to genomic medicine?



Variant calling

Manual review

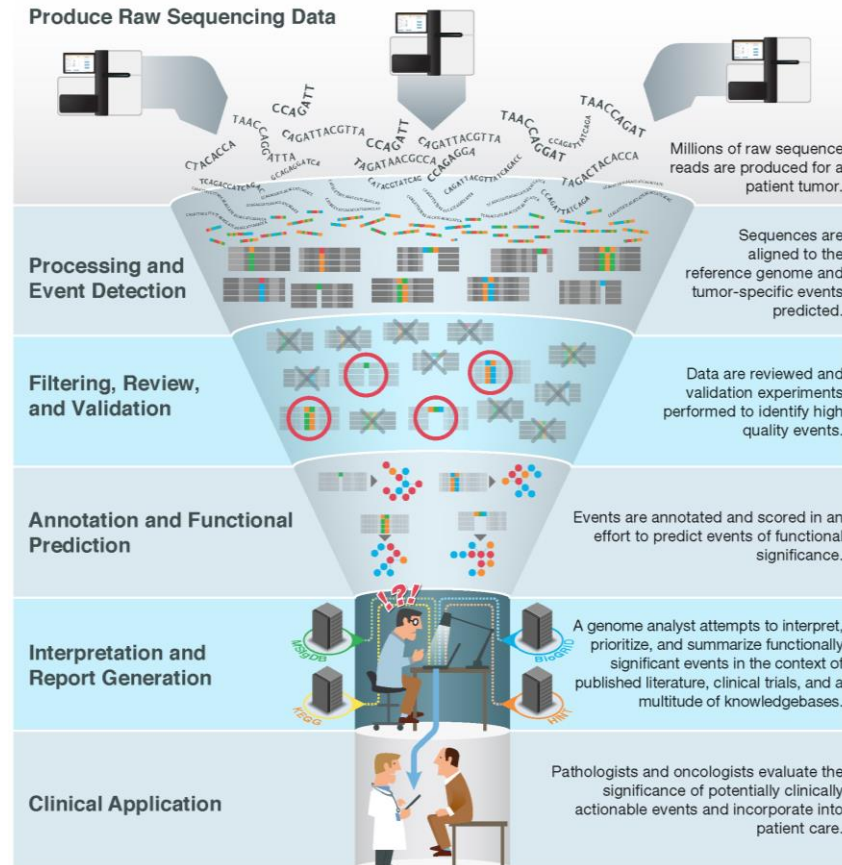
Functional prediction

Clinical Relevance

Decision-making



How else can AI be applied to genomic medicine?



Variant calling

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I) Manual review - how do we know a cancer variant is real?



Variant calling from raw sequence data is a critical step in most cancer genomics research and clinical laboratory genomics workflows

EGFR NM_005228

Protein length

COSMIC

7280 of 7375 mutations

34 tissue types

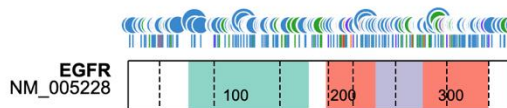
hg19 Pediatric COSMIC ClinVar CIViC

100 200

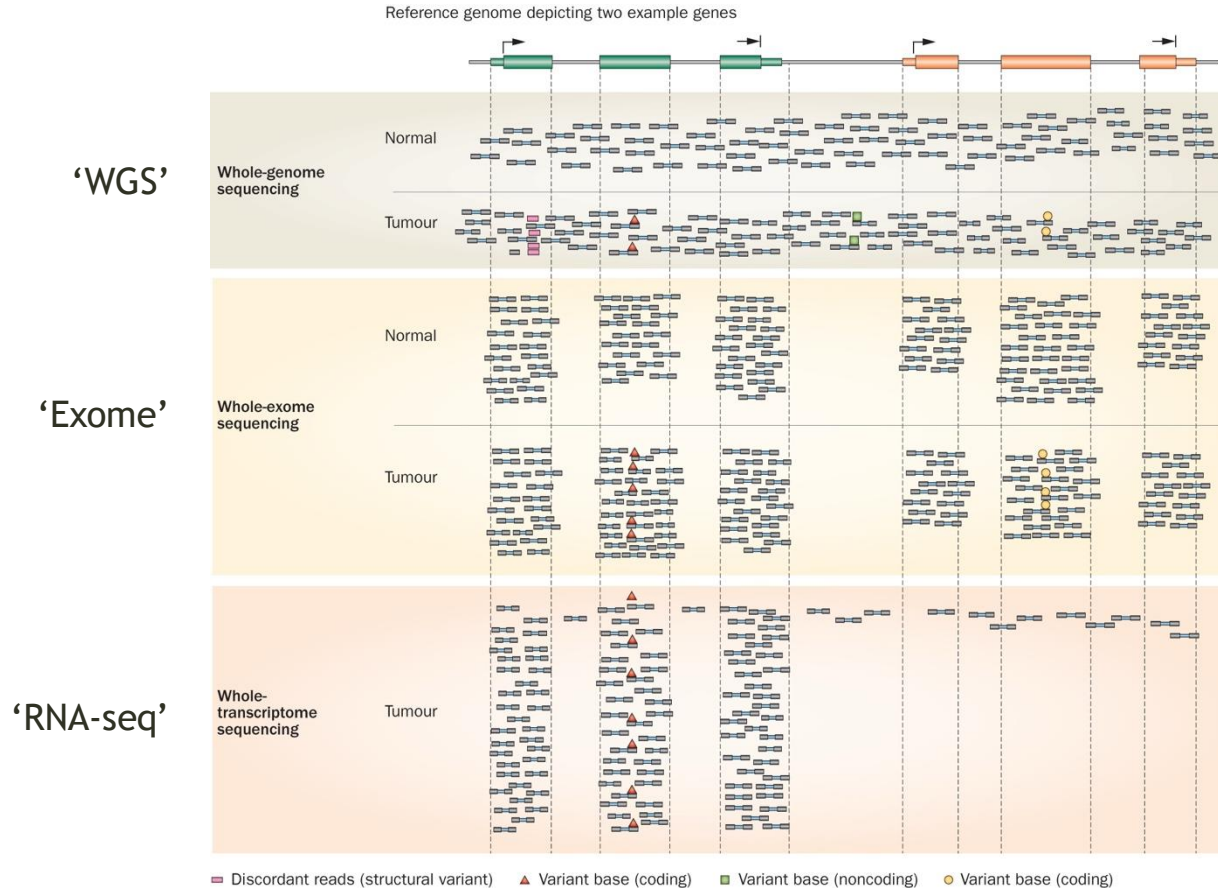
ABOUT

GENOMIC ALTERATIONS

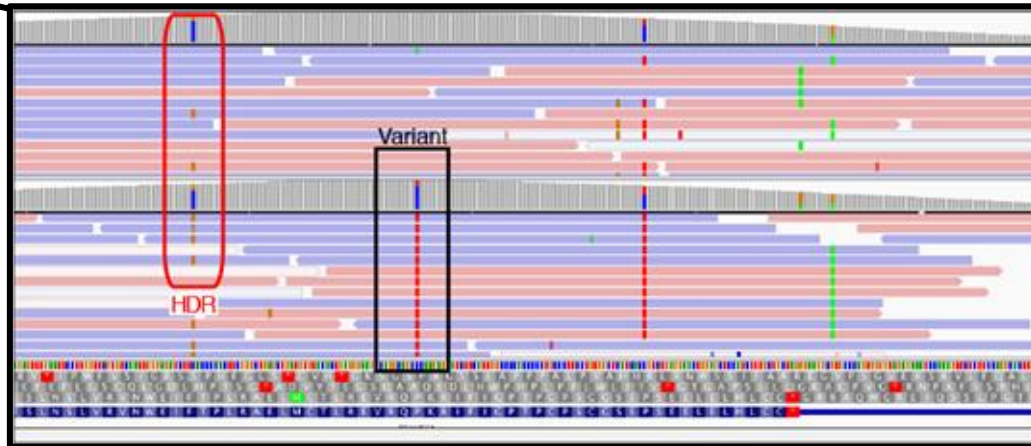
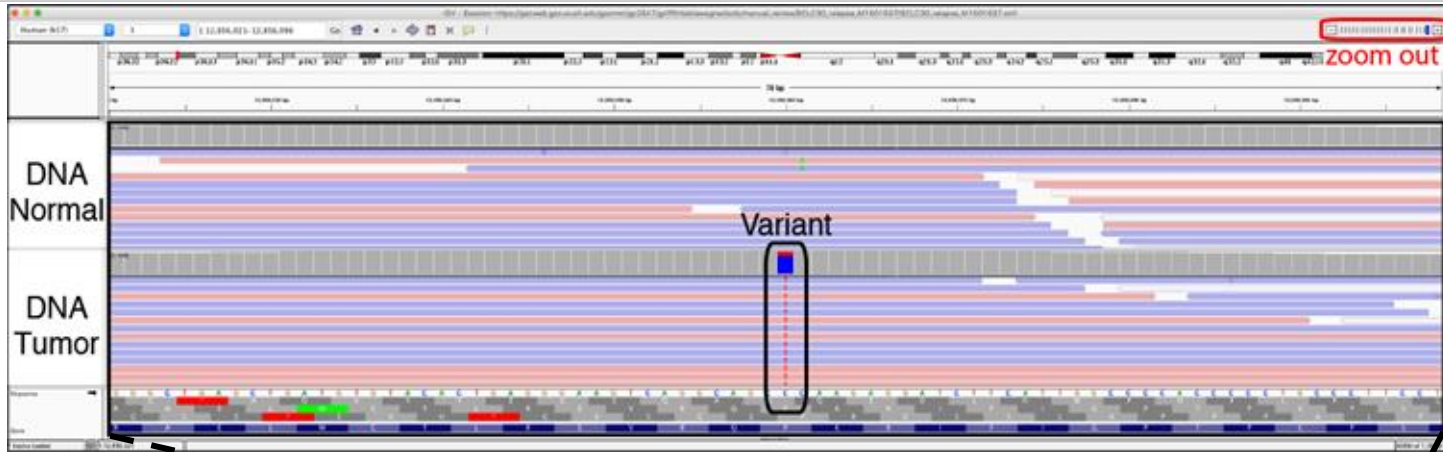
GENE ALTERATION	INTERPRETATION
PIK3CA H1047R	Mutations in PIK3CA have been reported in 26% to 33% of breast cancer cases (COSMIC, Jun 2012 and Kalinsky et al., 2009; 19671852). Activating mutations in PIK3CA, such as the one seen here, may predict sensitivity to inhibitors of PI3 kinase or its downstream signaling pathway (the PI3K/Akt/mTOR pathway) (Huang et al., 2007; 18079394). The mTOR inhibitors temsirolimus and everolimus have been tested in several clinical trials in breast cancer, and have been approved by the FDA for use in other tumor types. Inhibitors of PI3K and Akt are currently in clinical trials in breast cancer, alone or in combination with other therapies. PIK3CA mutations may play a role in resistance to hormonal therapy in ER+ breast cancers (Miller et al., 2011; 22114931). Activating mutations in PIK3CA may also confer resistance to anti-Her2 therapies (Chakrabarty et al., 2010; 20581867, Kataoka et al., 2010; 19633047, Wang et al., 2011; 21676217); combined inhibition of Her2 and the PI3K pathway may be required in tumors with ERBB2 amplification and PIK3CA mutation, though this remains an area of active investigation.
CCND1 amplification	CCND1 amplification has been reported in approximately 10-15% of invasive breast cancers, more frequently in BRCA-negative cancers (Elsheikh et al., 2008; 17653856, Bane et al., 2011; 21327470). There are no approved therapies that directly target the protein product of CCND1 (Cyclin D1); however, CCND1 amplification may predict sensitivity to inhibitors of Cdk4 and Cdk6, which are currently under investigation in clinical trials. Overexpression of Cyclin D1 has also been associated with resistance to endocrine therapy in breast cancer (reviewed in Lange et al., 2011; 21613412; Musgrove and Sutherland, 2009; 19701242, Butt et al., 2005; 16113099).
CDH1 E167*	CDH1 mutations are present in approximately 17% of breast cancers, and more often in luminal type cancers (COSMIC, Jun 2012, Hollestelle et al., 2010; 19593635). Loss of the E-cadherin protein, which is encoded by the CDH1 gene, has been associated with poor prognosis in triple negative breast cancer (Kashiwagi et al., 2010; 20551954, Tang et al., 2011; 21519872). Presently, there are no targeted therapies to address loss of CDH1/E-cadherin.



Whole genome, exome and transcriptome sequencing allows us to detect and confirm many different types of 'omic events



Manual review involves human inspection of sequence alignments to identify artifacts missed by automated variant callers



Literature is full of references to manual review and ad hoc filtering

“Mutations...were called with MuTect and filtered with oxidation and panel of normal samples filters to remove artefacts...”



Recurrent and functional regulatory mutations in breast cancer

Esther Kheradpour^{1,2}, Prasanna Parananan¹, Jonna Grimsby¹, Grace Xiao¹, Jesse M. Engreitz^{1,2}, Jangli Kim¹, Michael S. Lawrence^{1,2}, Arturo Taylor-Winters¹, Sergio Rodriguez-Corvea¹, Maria Rosenberg¹, Julian Hess¹, Chip Stewart¹, Yongli Ma¹, Marinka^{1,2}, Peter Stojanovic¹, Maria L. Cortez¹, Sara Segev¹, Carrie Chiklis¹, Adam Tracy¹, Trevor J. Pugh¹, Jesse Lee¹, Zongli Zheng¹, Lei W. Elston^{1,2}, A. John Iavarone¹, Jesse S. Boehm¹, Stacey B. Gabriel¹, Matthew Meyerson^{1,2}, Todd R. Golub^{1,2}, Jose Baselga¹, Alfredo Hidalgo-Minisola¹, Yoshi Shinda¹, Andre Bernards¹, Eric S. Lander^{1,2} & Gad Getz^{1,2,3,4}



An immunogenic personal neoantigen vaccine for patients with melanoma

Patrick A. Ott^{1,2,3,4}, Zhuoting Liu^{1,4}, Derin B. Jenkins^{1,2,3}, Sachet A. Shukla^{1,2,3}, Bing Sun¹, David J. Buzyr¹, Wendi Zhang¹, Adrienne Luciani¹, Anita Gubler-Harder¹, Lauren Peter^{1,2}, Christina Chen¹, Orkai Oliva¹, Todd A. Carter¹, Shuangping Li¹, David J. Little¹, Thomas Eisenhaue¹, Felicia Giger¹, Jonathan Swann¹, William J. Lane¹, Indu Jayaraj¹, Kalappanadar Nellaiappan¹, Andres M. Salazar¹, Heather Daley¹, Michael Seaman¹, Elizabeth I. Hachtbinder^{1,2,3}, Charles H. Yoon^{1,2,3}, Maegen Harder¹, Nali Lennon¹, Stacey Gabriel¹, Scott J. Rodig^{1,2}, Dan H. Bamach^{1,2,3}, Jon C. Aster^{1,2,3}, Gad Getz^{1,2,3,4}, Kai Wucherpfennig^{1,2}, Donna Neuberg¹, Jerome Ritz^{1,2,3}, Eric S. Lander^{1,2}, Edward F. Fritschy¹, Nir Hacohen^{1,2,3,4} & Catherine J. Wu^{1,2,3,4}

“All indels were manually reviewed in Integrative Genomics Viewer.”

“Coverage of at least 50 reads in both tumor and normal samples, >20% of reads supporting the variant in tumor samples, and <5% of reads supporting the variant in normal samples.”



RNF43 is frequently mutated in colorectal and endometrial cancers

Martin Glanville^{1,2,3,4}, Evan Hadji^{1,2,3,4}, Xinmeng Jia^{1,2,3,4}, Mei Yamashita¹, Joseph Rosenbluh^{1,2}, Kristian Cibulski¹, Gordon Sakuma¹, Michael S. Lawrence^{1,2}, Zhi Rong Qian¹, Reiko Nishihara^{1,2,3}, Eleanor M. Van Allen^{1,2}, William C. Hahn^{1,2,3}, Henry B. Gelboin¹, Eric S. Lander^{1,2,3,4}, Gad Getz^{1,2,3}, Shoji Ogino^{1,2,3,4}, Charles S. Fuchs^{1,2,3} & Levi A. Garraway^{1,2,3,4}

collected from participants in 2 prospective cohort studies, the Nurses' Health Study (NHS) and the Health Professionals Follow-up Study (HPFS), showed RNF43 mutations in 16 (16.9%) cases (median allelic fraction of 0.23, range of 0.01–0.60) (Supplementary Table 1). Frameshift mutations encoding p.Gly509S and p.Arg117fs constituting insertions or deletions of 1 bp in homopolymeric tracts (mononucleotide instability (MSI) loci) of seven and six C-to-G pairs, respectively, accounted for 41.7% (p.Gly509S) and 8.3% (p.Arg117fs) of the RNF43 mutations identified (Fig. 1a). To exclude the possibility that these mutations represented technical artifacts, we validated 51 of the RNF43 mutations (97% of the 52 mutations that had sufficient DNA available and achieved coverage of >50x in

- Rarely are the methods/criteria for manual review described
- Definitely not standardized

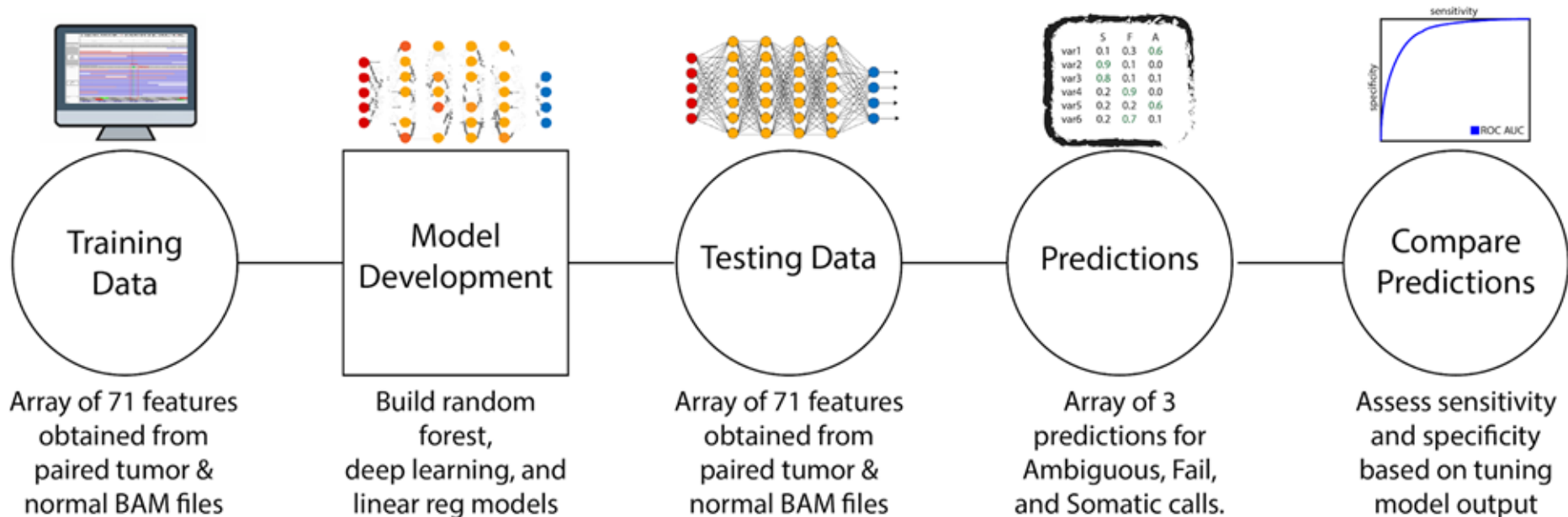


Manual review is a largely undocumented source of error, suffers from inter-reviewer variability, and requires significant resources

- We completed a study of independent/blinded manual review of a set of variants.
- High rates of inter-reviewer variability
 - Only "fair agreement" - Kappa = 0.37
 - 77.3% good (3/3 reviewers) or acceptable (2/3 reviewers) agreement on variant calls
- Manual review is labor-intensive, time-consuming
 - ~70-100 variants per hour



Can manual review be automated with AI?



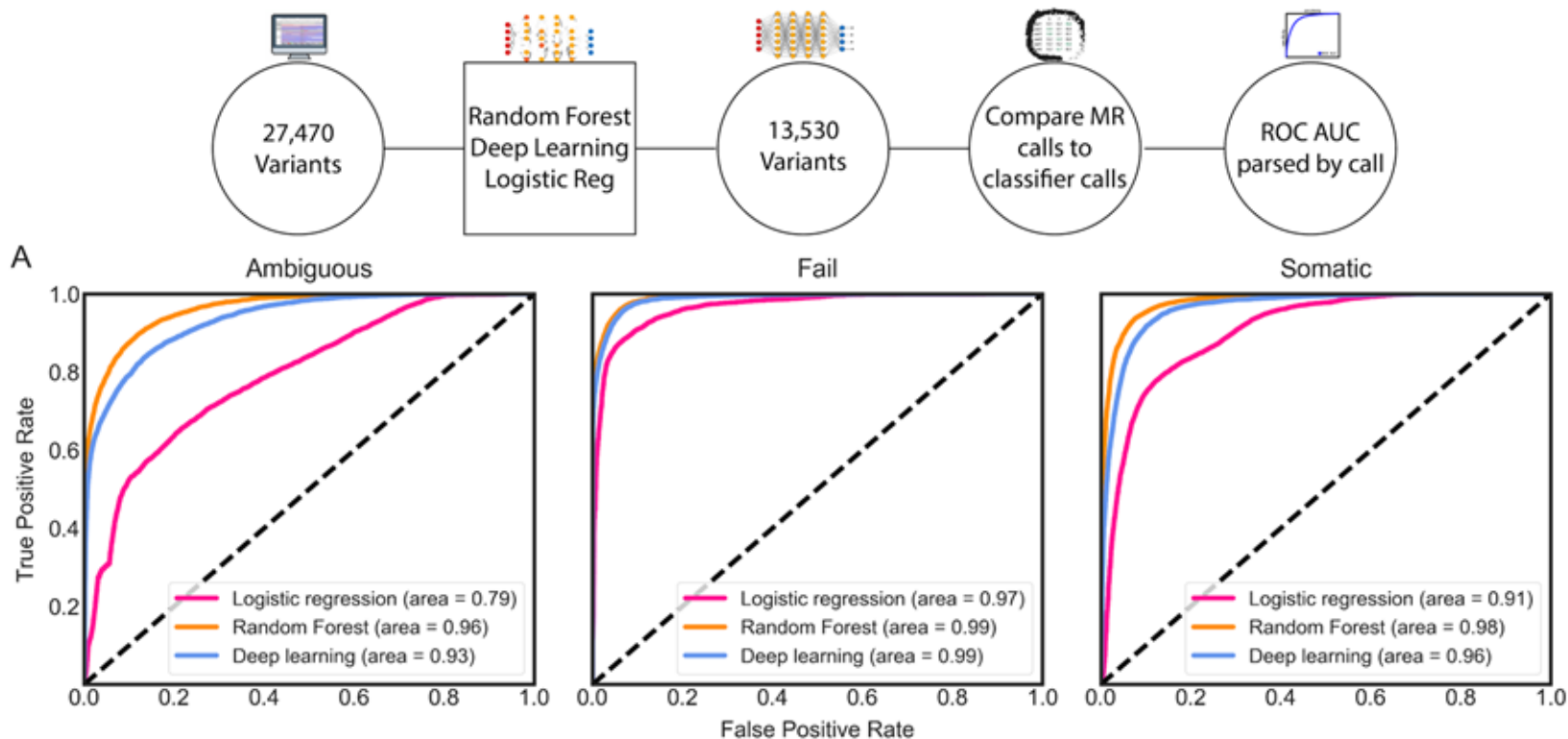
Assembled database of 40,000+ manually reviewed variants

	Training Set (n=27,470)	Testing Set (n=13,530)	Total (n=41,000)
Malignancy (410 cases)			
Leukemia (243 cases)	5,815	2,877	8,692
Lymphoma (23 cases)	1,263	628	1,891
Breast (135 cases)	8,986	4,320	13,306
Small Cell Lung (18 cases)	9,177	4,601	13,778
Glioblastoma (17 cases)	844	412	1,256
Melanoma (1 cases)	185	100	285
Colorectal (1 case)	842	419	1,261
Gastrointestinal Stromal (1 case)	70	31	101
Malignant Peripheral Nerve Sheath (1 case)	288	142	430
Sequencing Methods			
Capture Sequencing	9,479	4,755	14,234
Exome Sequencing	9,367	4,677	14,044
Whole Genome Sequencing	8,624	4,098	12,722
Variant Calls			
Somatic	12,266	6,115	18,381
Ambiguous	7,189	3,454	10,643
Fail	5,909	2,945	8,854
Germline	2,106	1,016	3,122

- Nine different tumor subtypes tested
 - 266 hematologic tumors and 144 solid tumors
- Multiple sequencing platforms tested
- Even distribution of variant calls (Somatic, Germline, Ambiguous, or Fail)
- Features used by manual reviewers extracted from alignments



Highly accurate (ROC AUC > 0.9) predictive model



- Random Forest and Deep Learning models out-perform Logistic Regression model
- Ambiguous calls are the most difficult to classify



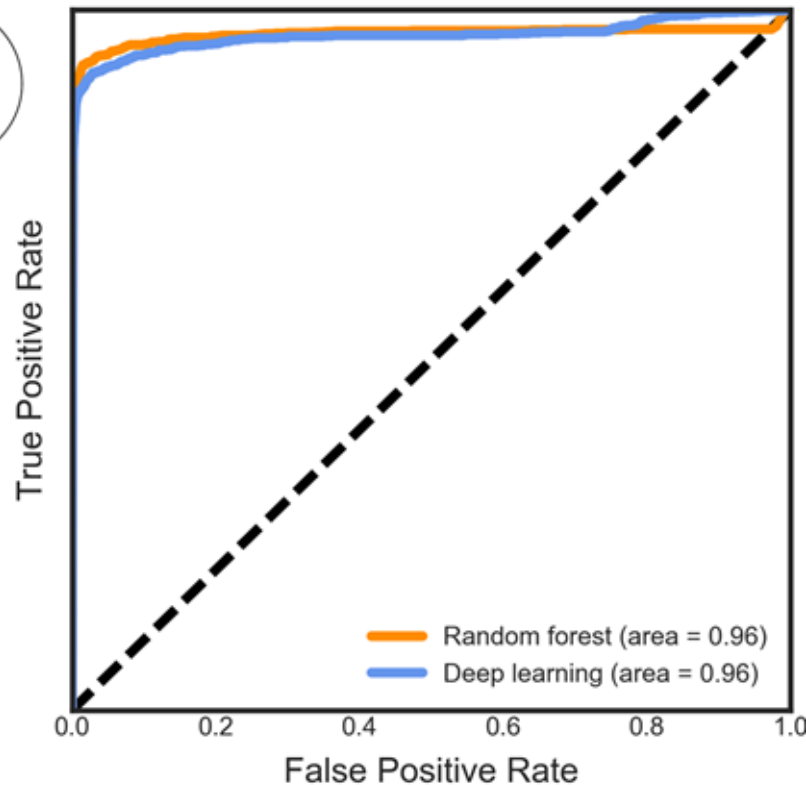
Excellent performance on Orthogonally validated variants



Orthogonal Validation Strategy:

- 192,241 variants were called by automated somatic variant callers.
- 1,343 true positives and 190,898 false positives were identified by 1000X sequencing
- Model predictions were compared to orthogonal sequencing results

Use of orthogonal sequencing data to classify variants performed exceptionally well



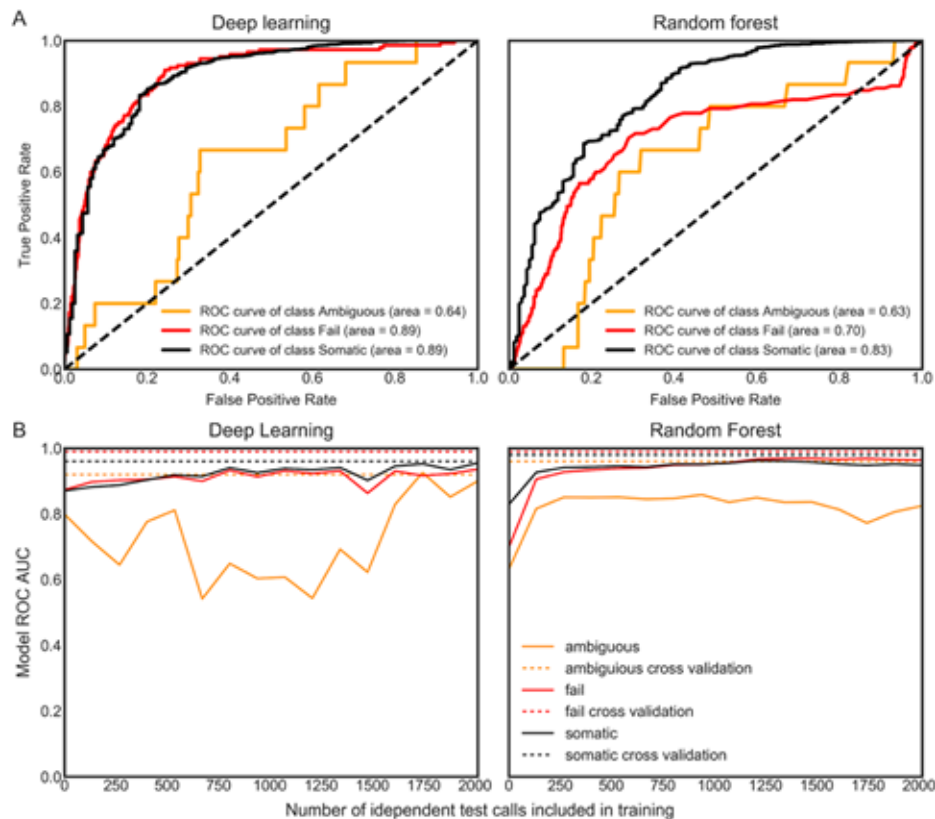
Performance is reduced on external datasets but can be recovered with a modest amount of re-training



“External Lab” Validation Strategy:

- Analyze performance on 4 SCLC cases with different manual reviewers, different sequencing platform, and different automated variant callers.
- Initial AUC showed suboptimal performance.
- Model was re-trained with 5% of manual review calls from new sequencing data.

Retraining the data with ~200 variants shows exceptional recovery of model performance.



A deep learning approach to automate refinement of somatic variant calling from cancer sequencing data

Benjamin J. Ainscough ^{1,2,12}, Erica K. Barnell ^{1,12}, Peter Ronning¹, Katie M. Campbell ¹, Alex H. Wagner ¹, Todd A. Fehniger ^{2,3}, Gavin P. Dunn⁴, Ravindra Uppaluri⁵, Ramaswamy Govindan^{2,3}, Thomas E. Rohan⁶, Malachi Griffith ^{1,2,3,7}, Elaine R. Mardis^{8,9}, S. Joshua Swamidass^{10,11*} and Obi L. Griffith ^{1,2,3,7*}

Cancer genomic analysis requires accurate identification of somatic variants in sequencing data. Manual review to refine somatic variant calls is required as a final step after automated processing. However, manual variant refinement is time-consuming, costly, poorly standardized, and non-reproducible. Here, we systematized and standardized somatic variant refinement using a machine learning approach. The final model incorporates 41,000 variants from 440 sequencing cases. This model accurately recapitulated manual refinement labels for three independent testing sets (13,579 variants) and accurately predicted somatic variants confirmed by orthogonal validation sequencing data (212,158 variants). The model improves on manual somatic refinement by reducing bias on calls otherwise subject to high inter-reviewer variability.



Benjamin
Ainscough



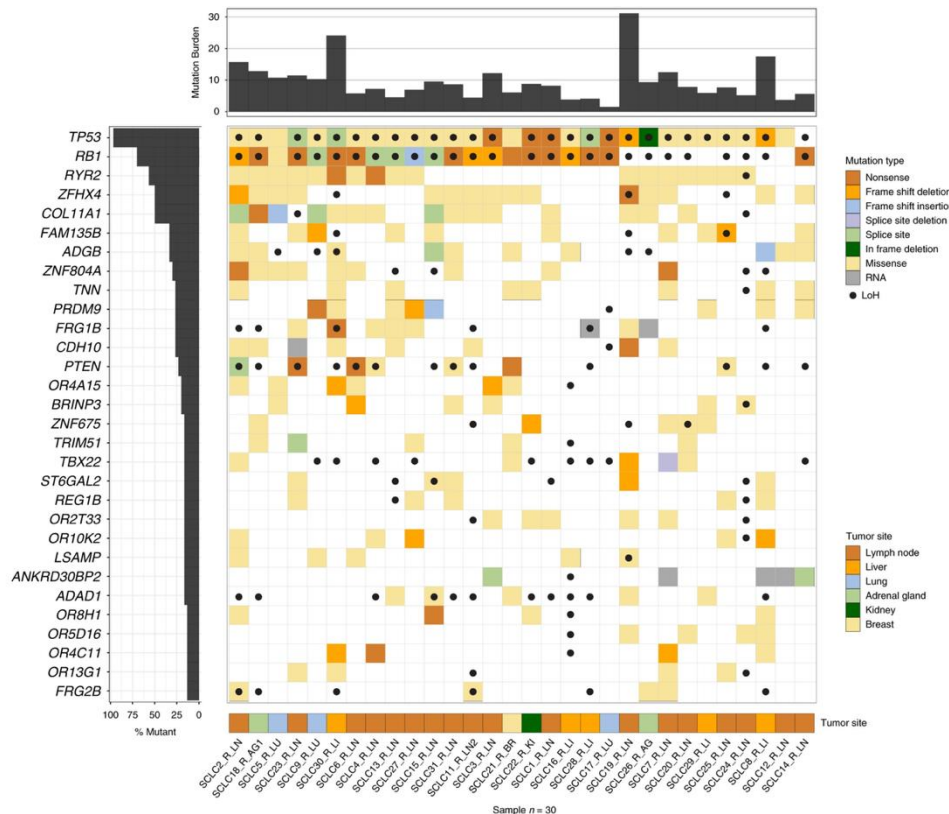
Josh
Swamidass



Erica
Barnell



We have applied this model to multiple large-scale cancer genome projects, including our survey of relapsed SCLC



- High mutation burden requiring review of 100s or 1000s of variants
- Confirmed TP53 and RB1 nearly universally co-altered in SCLC
- COL11A1 next most significantly mutated, may mediate resistance to platinum chemotherapy
- Relapse samples also characterized by recurrent alterations in WNT signaling



ARTICLE

DOI: 10.1038/s41467-018-06162-9

OPEN

Recurrent WNT pathway alterations are frequent in relapsed small cell lung cancer

Alex H. Wagner¹ , Siddhartha Devarakonda^{2,3}, Zachary L. Skidmore¹, Kilannin Krysiak^{1,2} , Avinash Ramu¹ , Lee Trani¹, Jason Kunisaki¹, Ashiq Masood^{2,3,9}, Saiama N. Waqar^{2,3}, Nicholas C. Spies¹, Daniel Morgensztern^{2,3}, Jason Waligorski¹, Jennifer Ponce¹, Robert S. Fulton¹, Leonard B. Maggi Jr.^{2,3,4}, Jason D. Weber^{2,3,4}, Mark A. Watson³, Christopher J. O'Connor⁵, Jon H. Ritter⁵, Rachelle R. Olsen⁶, Haixia Cheng⁶, Anandaroop Mukhopadhyay⁶, Ismail Can⁶ , Melissa H. Cessna⁷, Trudy G. Oliver⁶, Elaine R. Mardis^{1,3,8,10}, Richard K. Wilson^{1,3,8,10}, Malachi Griffith^{1,2,3,8}, Obi L. Griffith^{1,2,3,8}  & Ramaswamy Govindan^{2,3}



Alex
Wagner



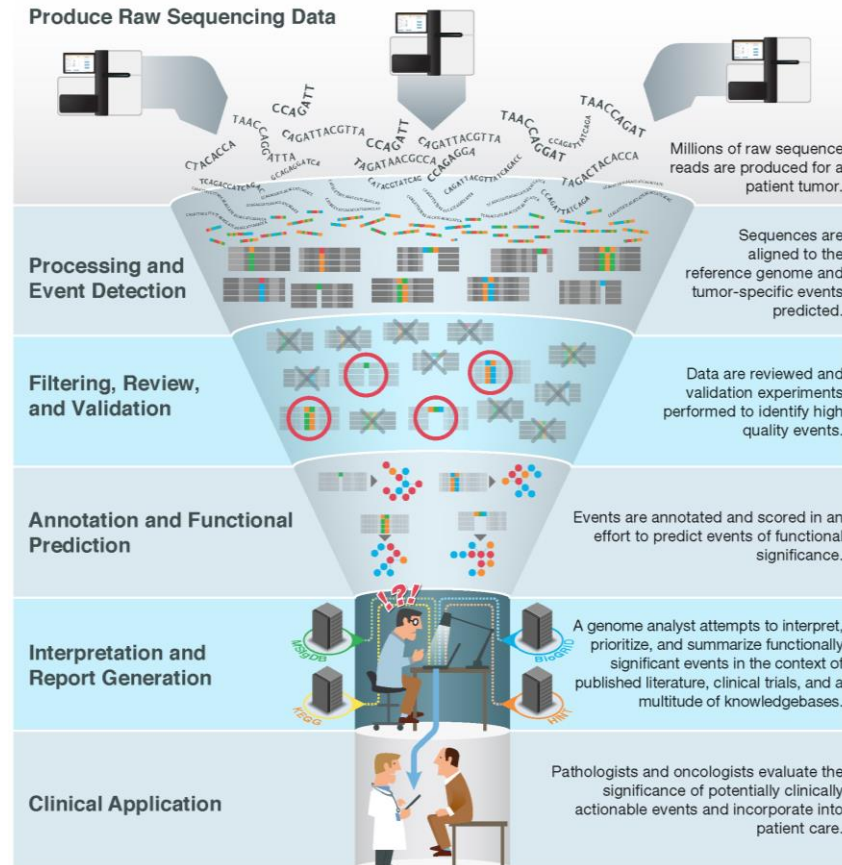
Ramaswamy
Govindan



Siddhartha
Devarakonda



How else can AI be applied to genomic medicine?



Variant calling

Manual review

Functional prediction

Interpretation

Decision-making



II) Interpretation - how do we know if a cancer variant is clinically significant?

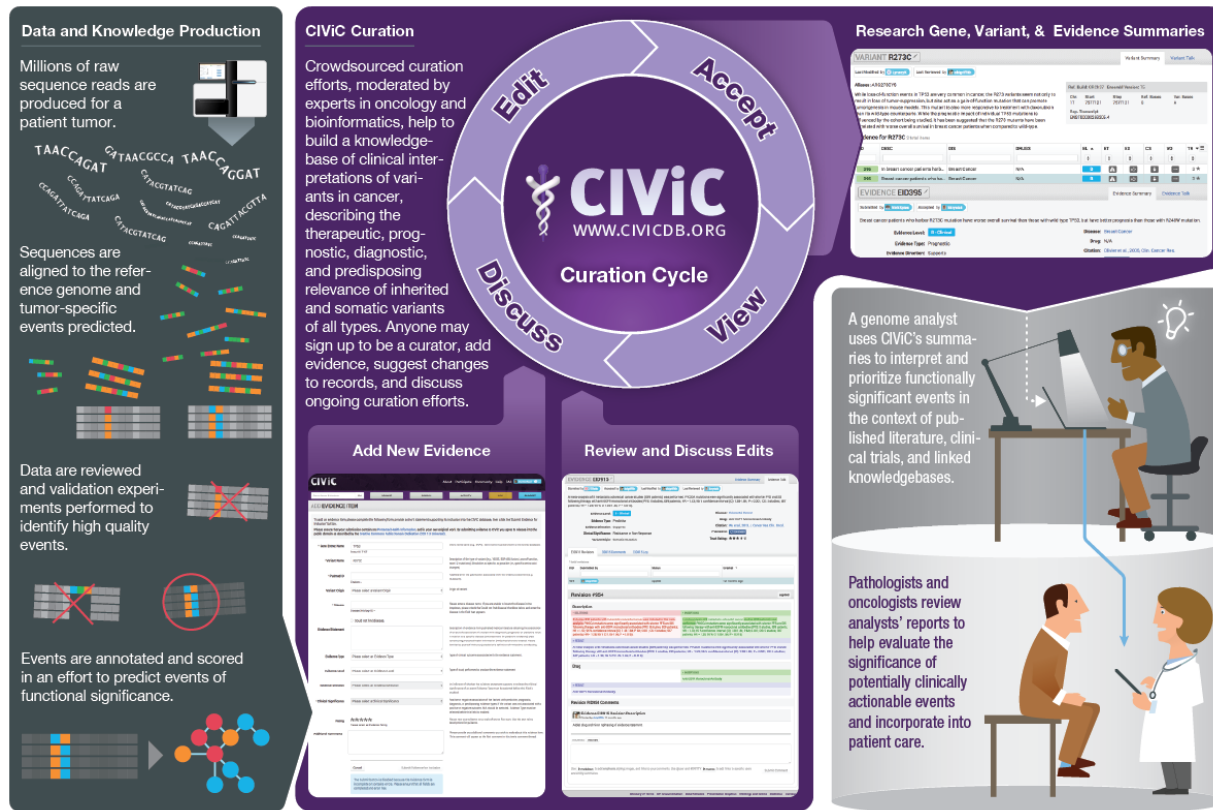


What is a variant of *significance* in cancer?

- **Predictive** of therapeutic response
 - *BRAF V600E predicts sensitivity to vemurafenib*
- **Diagnostic** of tumor subtype
 - *DNAJB1-PRKACA fusion differentiates fibrolamellar hepatocellular carcinoma from conventional HCC*
- **Prognostic** of survival change
 - *TP53 mutations are associated with worse progression-free survival in lung adenocarcinoma*
- **Predisposing** for cancer development
 - *Patients with the RUNX1 Y260* mutation are associated with increased risk of developing acute myeloid leukemia*



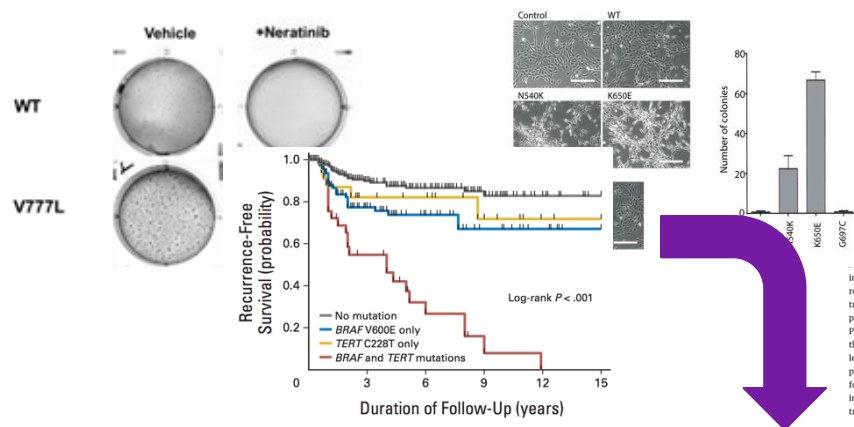
CIViC: an open knowledgebase and curation system for clinical interpretation of variants in cancer



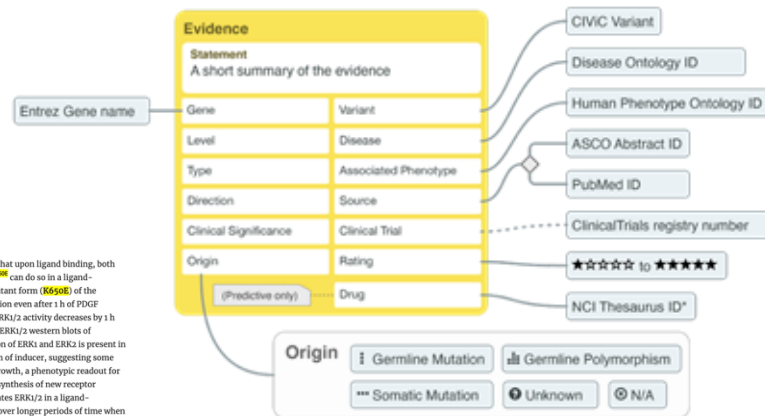
Heterogeneous evidence

->

Structured data model



K650E mutation. We have previously shown that upon ligand binding, both wild-type and mutant EGFR activate ERK1/2 and that EGFR can do so in a ligand-independent manner (33). Here, we find that the mutant form (K650E) of the receptor produces a highly sustained ERK1/2 activation even after 1 h of PDGF treatment, whereas in wild-type expressing cells, ERK1/2 activity decreases by 1 h post-PDGF addition (Fig. 3A). Using anti-phospho-ERK1/2 western blots of PDGF-stimulated cell lysates, phosphorylation and activation of ERK1 and ERK2 is present in the mutant cell at low levels even before the addition of inducer, suggesting some leakiness within the system. However, neurite outgrowth, a phenotypic readout for productive receptor activity, is only observed upon synthesis of new receptor following induction. Therefore mutant EGFR3 activates ERK1/2 in a ligand-independent manner and maintains this activation over longer periods of time when treated with ligand than does the wild-type receptor.



CIVIC
CLINICAL INTERPRETATION OF
VARIANTS IN CANCER

Quicksearch

Home About CIVIC Help

Evidence / EID288 / Summary

EID288 Parents: ER882 V777L

Revise Flag

Curators: Editors:

EID288

Summary Comments Revisions Flags Events

Description

In MCF10A cell lines, the V777L mutation was shown to be sensitive to neratinib.

Type Predictive **Direction** **Supports**

Clinical Significance Sensitivity / Response

Level D **Rating** ★★★★★

Status Accepted **Submitted** Jun 21, 2015 by NickSpies **Accepted** Jun 21, 2015 by kkrysiak

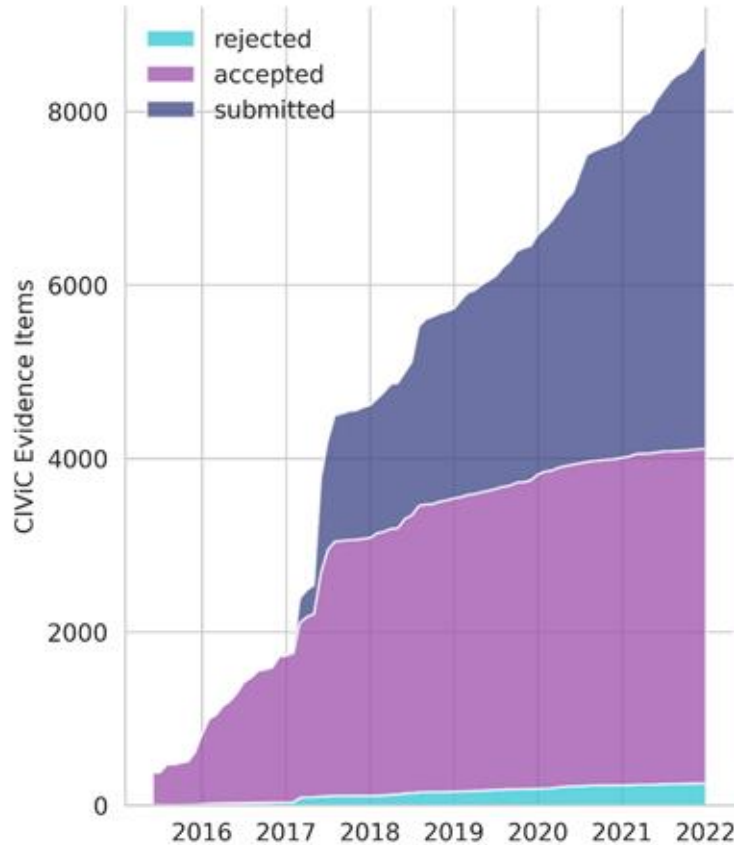
Source PubMed: Bose et al., 2013, Cancer Discov

Clinical Trial None Specified

Disease Breast Cancer **Phenotype** None Specified

Drug Neratinib

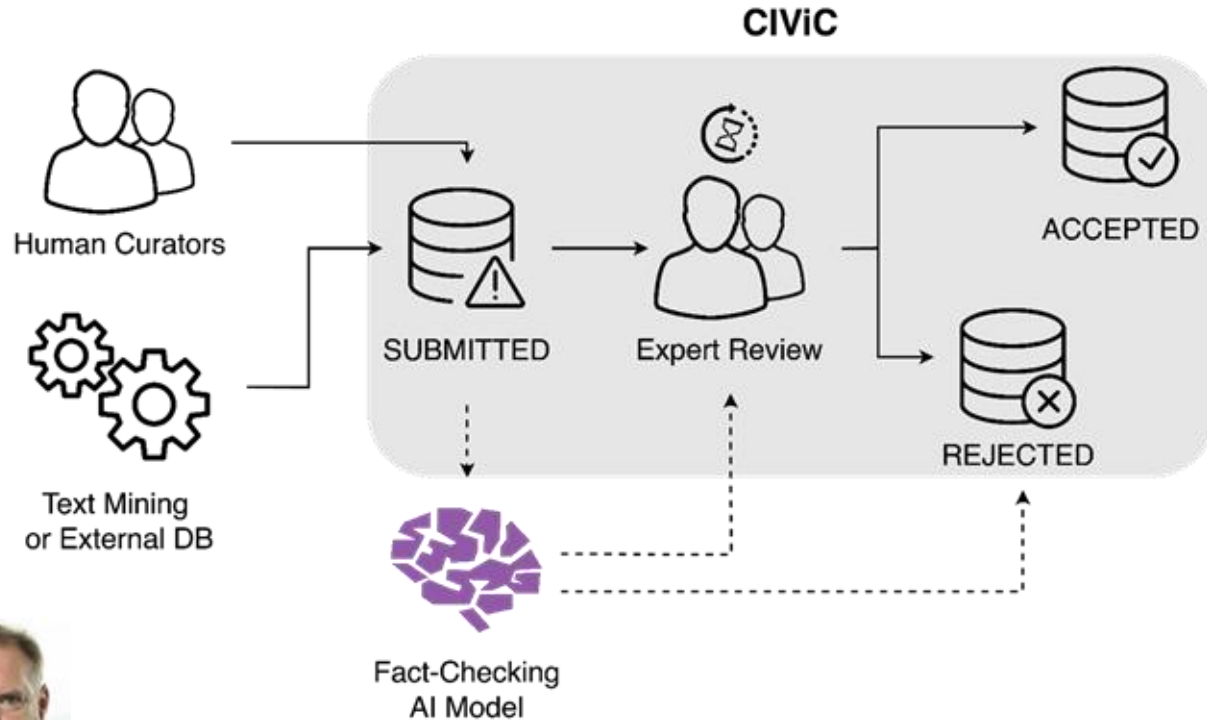
Expert moderation of submitted content is a major bottleneck



- Entries are being submitted to CIViC faster than experts can review them (right)
- Automated fact-checking could alleviate some of this burden
- Dearth of open-data scientific datasets in fact-checking and natural language interpretation



Integration of Fact Checking AI into CIViC



Caralyn Reisle (Steven Jones)



Annotation Process: Creating the Cancer-Facts Dataset

downstream effectors in wild-type or mutated ALK-expressing BaF3 cells depleted of IL-3 for 6 hours. The mobilities of molecular weight (MW) standards are shown on the left. *a*, Growth of mutated ALK-expressing BaF3 cells exposed to TAE684 for 72 hours. The values are means $\pm$ SD of triplicate experiments.

The BaF3 assay has been validated for a broad spectrum of oncogenic tyrosine kinase alleles including mutant EGFR^{L858R} and FLT3^{ITD}, and thus we treated BaF3 cells expressing each of the ALK mutations with increasing concentrations of TAE684, a highly potent ALK inhibitor^{4,12,13}. The activating mutation, F1174L, was found to be extremely sensitive to TAE684, with an IC₅₀ of 8 nM, identical to that of NPM-ALK-expressing BaF3 cells. The R1275Q mutation was also sensitive to TAE684, albeit with a much higher IC₅₀ of 328 nM. In contrast, BaF3 cells expressing FLT3-ITD or wild-type ALK, did not respond to TAE684 (IC₅₀ 4.5 μ M; Fig. 1*a*).

Analysis of the ALK gene in a panel of 30 neuroblastoma cell lines revealed sequence variants in 6, including 3 different cell lines containing the F1174L mutation (Kelly, SH-SY5Y and LAN-1), which was also the most common mutation in the primary tumours (Table 1). An R1275Q mutation, identical to the one found in primary sample 411, was also detected in the SMS-KCNR cell line. We observed dose-dependent growth inhibition of the SH-SY5Y (F1174L) and Kelly (F1174L) neuroblastoma cell lines with increasing concentrations of TAE684 (IC₅₀ of 258 and 416 nM respectively; Fig. 2*a*). These results are

Evidence is selected using
hypothes.is

Hypothes.is Selections are Displayed in
CIViC via a Custom Chrome Extension

Claim/Evidence Pair Labels are Confirmed
with a 2nd Round of Annotation



Evidence Item Consistency Analysis

Significance	♥ Sensitivity / Response
Type	👁 Predictive
Molecular Profile	🟢 MGMT Promoter Methylation
Disease	🔴 Glioblastoma
Phenotype	<i>Not specified</i>
Therapy	🟣 Temozolomide

Prompt

Given a claim and some evidence, determine the stance of the evidence toward the claim. Use the following labels: SUPPORTS, REFUTES, and NEI (not enough information). Give your reasoning.

Statement	
In a randomized clinical trial, patients with MGMT promoter methylation benefitted from temozolomide. This benefit was also methylation status dependent, as those without methylation did not see increased survival.	
Direction	👍 Supports

Expected Response

claim: molecular profile is <MP>; evidence type is <evidence type>; significance is <significance>; therapy is <therapy>; disease is <disease>

evidence: <statement>

label: <direction>



Example #1: AI fact-checking model flags errors in evidence direction field

EID7393 Summary Comments Revisions Flags Events Revise Flag

Curators:

Statement

In the patient-derived cell line GIST430 harboring KIT V560_L576del primary activating mutation implanted in mice, imatinib and ponatinib were shown to reduce KIT phosphorylation compared to the vehicle. In GIST430 cell lines, ponatinib and imatinib reduced phospho-AKT and phospho-ERK levels. A cell line derived from GIST430 patients expressing KIT V560_L576del primary mutation demonstrated sensitivity to imatinib (IC50: 61nmol/L) and ponatinib (IC50: 12nmol/L) treatments. In comparison, the insensitive KIT-independent GIST226 cell line showed IC50 >5000nmol/L and 2807nmol/L for these drugs respectively. IC50 was determined by assessing cell viability.

Type Predictive **Direction** Does Not Support

Significance Sensitivity / Response **Variant Origin** Somatic

Level Preclinical evidence **Rating** ★★★★☆

Status Accepted **Submitted** (Jul 8, 2019) **Accepted** (Jul 9, 2019) by BrianLi by ArpadDor

Molecular Profile KIT V560_L576DEL

MP Expression KIT V560_L576DEL

Disease Gastrointestinal Stromal Tumor

Phenotype Not specified

Therapies Ponatinib Imatinib

Therapy Interaction Type Substitutes

[mosaicml__mpt-7b-chat]: **SUPPORTS**. The evidence suggests that both imatinib and ponatinib effectively target GIST molecular profile with high sensitivity . . .

[BioMistral__BioMistral-7B-DARE]: **SUPPORTS**. The cell line derived from GIST430 patients expressing KIT V560_L576del primary mutation demonstrated sensitivity to imatinib and ponatinib treatments . . .

▼ **RID60338 Evidence Direction Updated** By kkrysiak for The McDonnell G

Original Evidence Direction **Evidence Direction Diff**

Does Not Support Supports



AI fact-checking model flags errors in significance field

EID1516

SummaryCommentsRevisionsFlagsEvents

Curators: [User Icons]

Statement

Patients with a triple mutation of NPM1, DNMT3A, and NRAS (G12/13) (three-way interaction $p=0.04$) showed a relatively benign prognosis with an 8-year survival rate of 75% and a 10-year survival rate of 55%. Conversely, patients with a concomitant NPM1/DNMT3A co-mutation, in the absence of an NRAS (G12/13) mutation, showed a clinically unfavorable 10-year survival rate of 30%.

Type

Prognostic

Direction

Supports

Significance

Poor Outcome

Variant Origin

Somatic

Level

B - Clinical evidence

Rating

★★★★☆

Source

PubMed: Papaemmanuil et al., 2016

Clinical Trial

NCT00146120

Status

Accepted

Submitted (Jul 7, 2016)

Accepted (Jul 7, 2016)

by EricaBarnell

by NickSpies

Molecular Profile

NRAS G12/G13

MP Expression

NRAS G12/G13

Disease

Acute Myeloid Leukemia

Phenotype

Not specified

Therapy

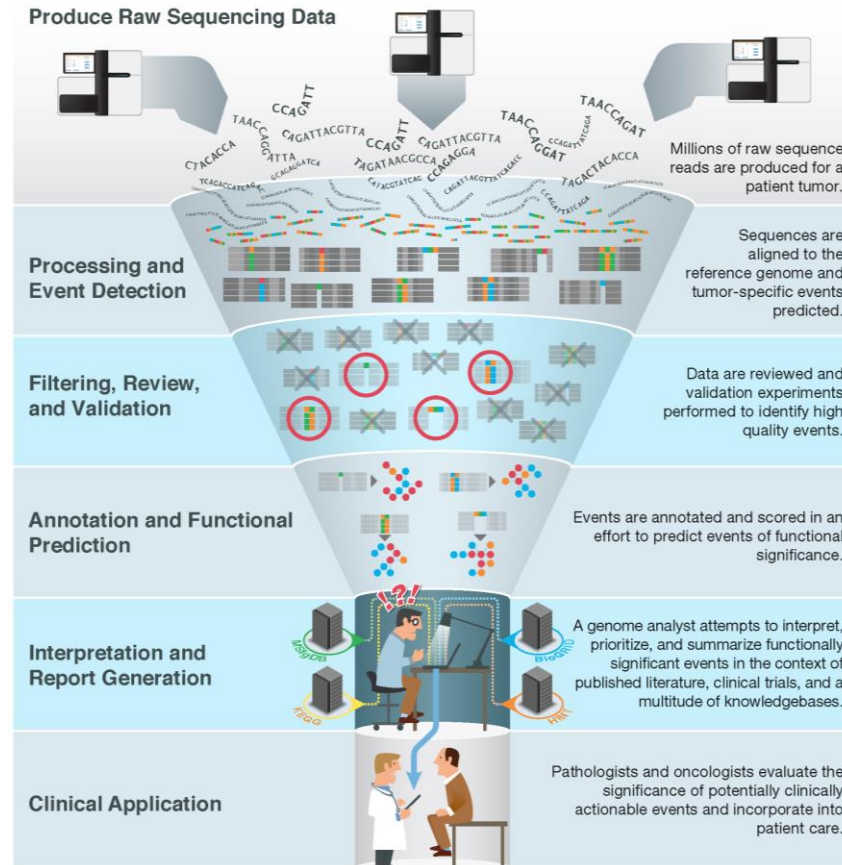
Not applicable



[BioMistral__BioMistral-7B-DARE]: All elements are present. Patients with NRAS G12/13 mutation had better outcome (8 yr OS 75%) than those without this mutation (10 yr OS 30%) . . .



How else can AI be applied to genomic medicine?



Communication

- figures
- grants
- letters
- homework

Variant calling

Manual review

Functional prediction

Interpretation

Decision-making



III) LLMs can help with a lot!

ChatGPT o1

Yesterday

Missouri Doc Fees Kia EV9

St Louis County Vehicle Tax

How old am I

Previous 7 Days

Clinical Trial Overview Figure

Cytogenetic region terminology

BRCA Mutations and PARP Inhi

Equation solving with Python

ctDNA vs cfDNA

Credit Check and Freeze

Interactive Bash Docker Sessio

Freeze Child Credit History

AI Growth Reports

How to Solve Rubik's Cube

Previous 30 Days

Cancer Etiology Overview

HSA Excess Contribution Earni

Temporary



What can I help with?

Ask anything



Search



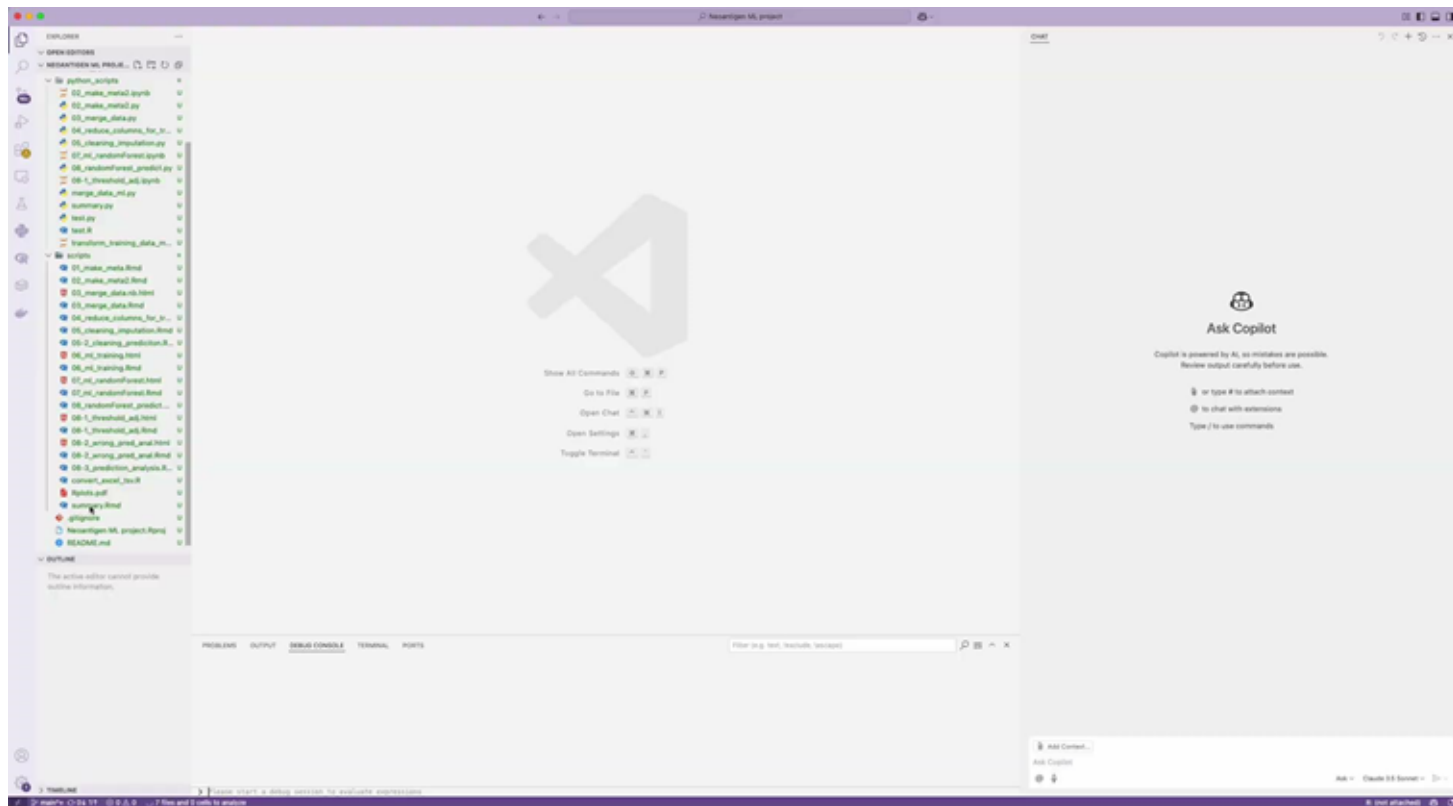
Deep research



Why am I trying to do this without using LLMs?



Convert an existing R ML model into Python equivalent



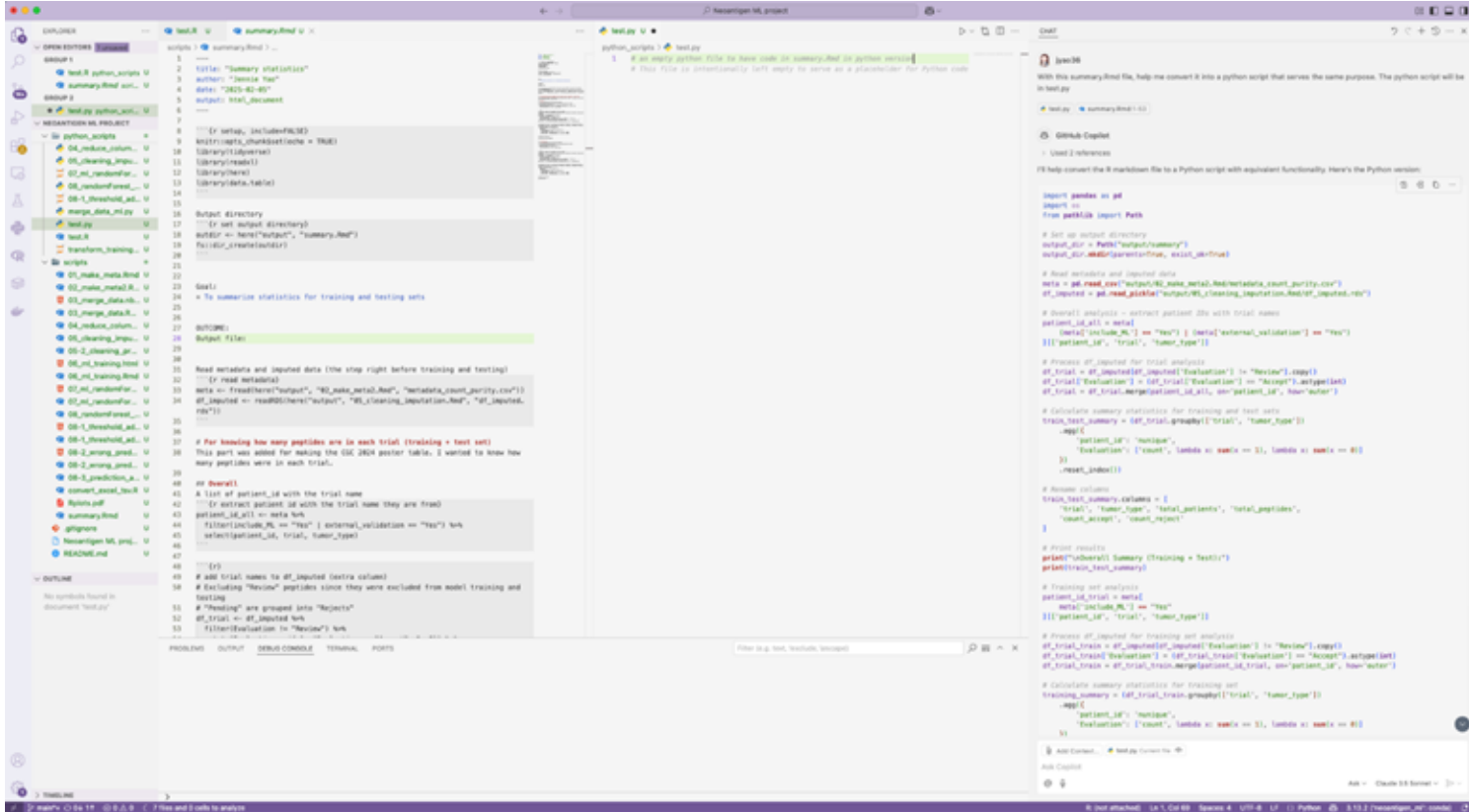
“Convert this Rmd file into a python script that serves the same purpose”



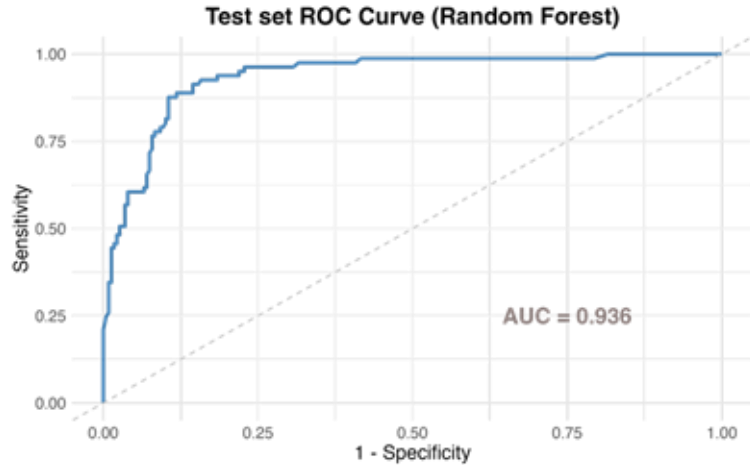
Jennie Yao



Convert an existing R ML model into Python equivalent



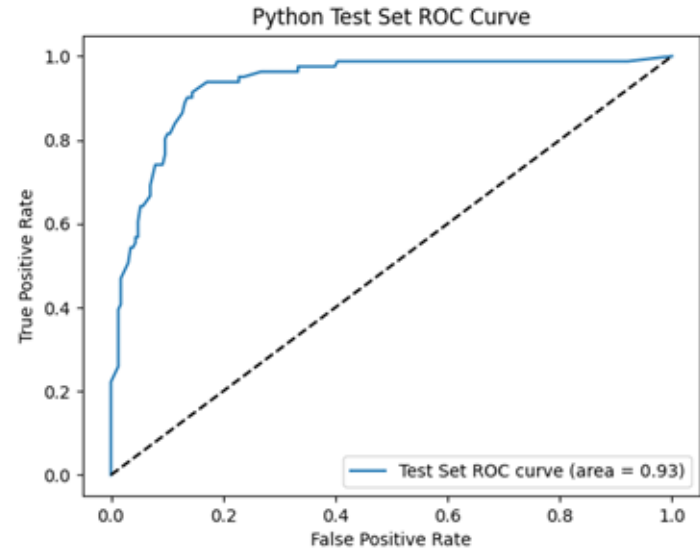
Test set performance concordant



R model performance:

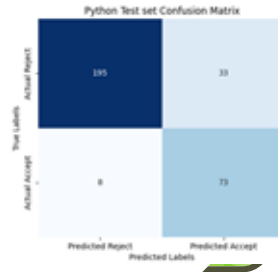
- 88.89% sensitivity
- 86.84% specificity
- 0.936 AUC

actual_values	rfd_s_pred	
	Reject	Accept
Reject	198	30
Accept	9	72



Python model test set performance:

- 90.12% sensitivity
- 85.52% specificity
- 0.932 AUC



Review consent forms for genomic data sharing language

I need to complete a table which documents whether a large number of informed consent forms (ICF) have language allowing the sharing of genomic/genetic data. This needs to be done for 60+ pdf forms. Specifically, I need to create a spreadsheet with the following four components (columns) for each form:

1. Genomic data sharing allowed? (Y/N)
2. If Genomic Data Sharing was Y, extract text from the ICF; otherwise, indicate N/A
3. Page number(s) in ICF detailing genomic data sharing language, formatted as “page(s) x-x”. Check to make sure only the page or pages from which the text is extracted are listed.
4. Name of consent file PDF

Here is an example of appropriate language: “These studies may include genetic research. Genetic research can look at DNA to look for mutation (changes) that may increase the risk of disease or affect the way a person responds to treatment. This could include the database of Genotypes and Phenotypes (dbGaP), which is a repository that shares genomic data. The data within dbGaP remains anonymous, but de-identification of data cannot eliminate a risk of disclosure, which could affect you and even your family members. ... We will share your blood, tissue, and data with other researchers. They may be doing research in areas similar to this research or in other unrelated areas. These researchers may be at Washington University, at other research centers and institutions, or industry sponsors of research. We may also share your research data with large data repositories (a repository is a database of information) for broad sharing with the research community. If your individual research data is placed in one of these repositories only qualified researchers, who have received prior approval from individuals that monitor the use of the data, will be able to look at your information”

The relevant language can be contiguous or spread across multiple parts of the document but must have enough surrounding text or headers to ensure that the meaning is clear and not taken out of context.

The language should be the same or similar to the examples above. The most critical question is whether the consent form made it clear that genetic/ genomic data would be shared with other researchers. However, if the consent form provides explanation of the genetic research to be performed, the use of specific genetic data repositories (e.g., dbGaP), access restrictions on such repositories, possible risks associated with sharing genetic data, or other aspects related to genetic research/data then that language can be extracted as well.

Keep the extracted result to a minimum span of text that covers genetic/genomic data and sharing of that data without including much language that more generally covers the sharing of blood, tissue, specimens or other kinds of data unless such language is needed to document implicitly allowing genetic data sharing. If there are multiple spans of text that accomplish the same goal, one good representative is sufficient. It is critically important that the language identified and extracted be quoted verbatim in the table and accurately linked to the page number(s) from which it came.

Create a table in excel format for output.

I will upload batches of PDFs from which to build the table.



Review consent forms for genomic data sharing language

Understood — I'm ready to help. Once you upload a batch of PDFs, I will process each one and generate a table with the following columns for each consent form:

1. Genomic data sharing allowed? (Y/N)
2. Extracted Text (verbatim, if Y; N/A if N)
3. Page number(s) with relevant text, formatted as "page(s) x-x"
4. Name of consent file PDF

I'll ensure:

- Only the page(s) containing the extracted text are listed.
- Text is quoted exactly from the source.
- Only relevant, appropriately contextual excerpts are included.

Please go ahead and upload your first batch of PDFs.



Here is the next batch of PDFs. Please analyze all 4 additional PDFs



Genomic Data Sharing Consent Table			
	Genomic Data Sharing	Extracted Text	PDF File Name
1	Y	We may also share your research data with large data repositories (a repository is a	202307205 mal
2	Y	We may also share yc These future studies These studies may in- These studies may in-	202307205 scri

All four additional consent PDFs have been analyzed and added to the table. Each one contains appropriate language explicitly allowing for the sharing of genomic data.

[Download new batch genomic data sharing summary](#)

Let me know if you'd like to continue with additional files!

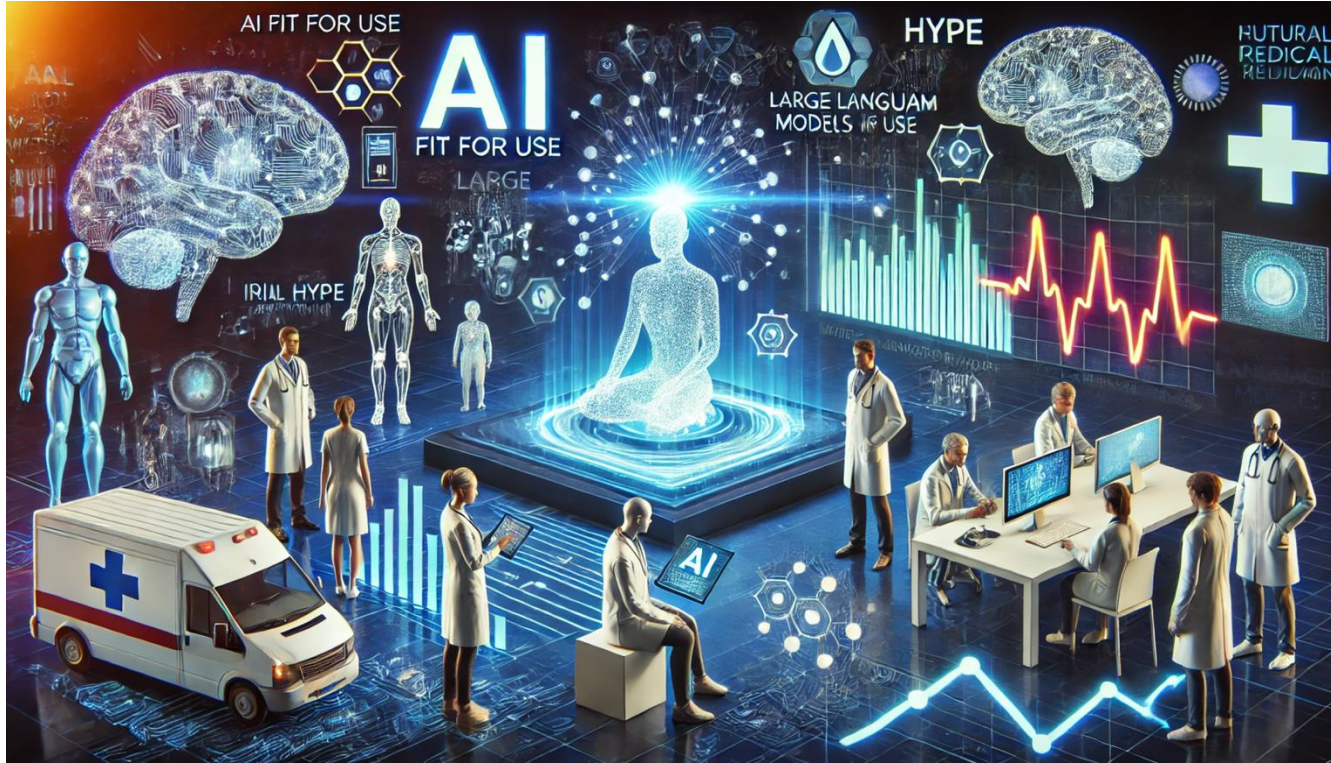


Make a figure for a review paper

I would like to create a high-level conceptual image to summarize a paper I wrote. The paper is titled "Title: Looking Forward to AI and Medicine: Where Are We, and Where Are We Going?" The key themes of the paper are: (1) Why this moment of AI hype could be different. The emergence of LLMs; (2) The importance of Fit for use - why this should be evaluated differently; (3) The need for evaluation - how it is different in medicine (data sharing, regulation, etc); (4) What we expect to happen - After initial hype, we enter a period of diminishing returns in the short/medium term but then improvements accelerate eventually leading to transformation of many areas of medical practice.



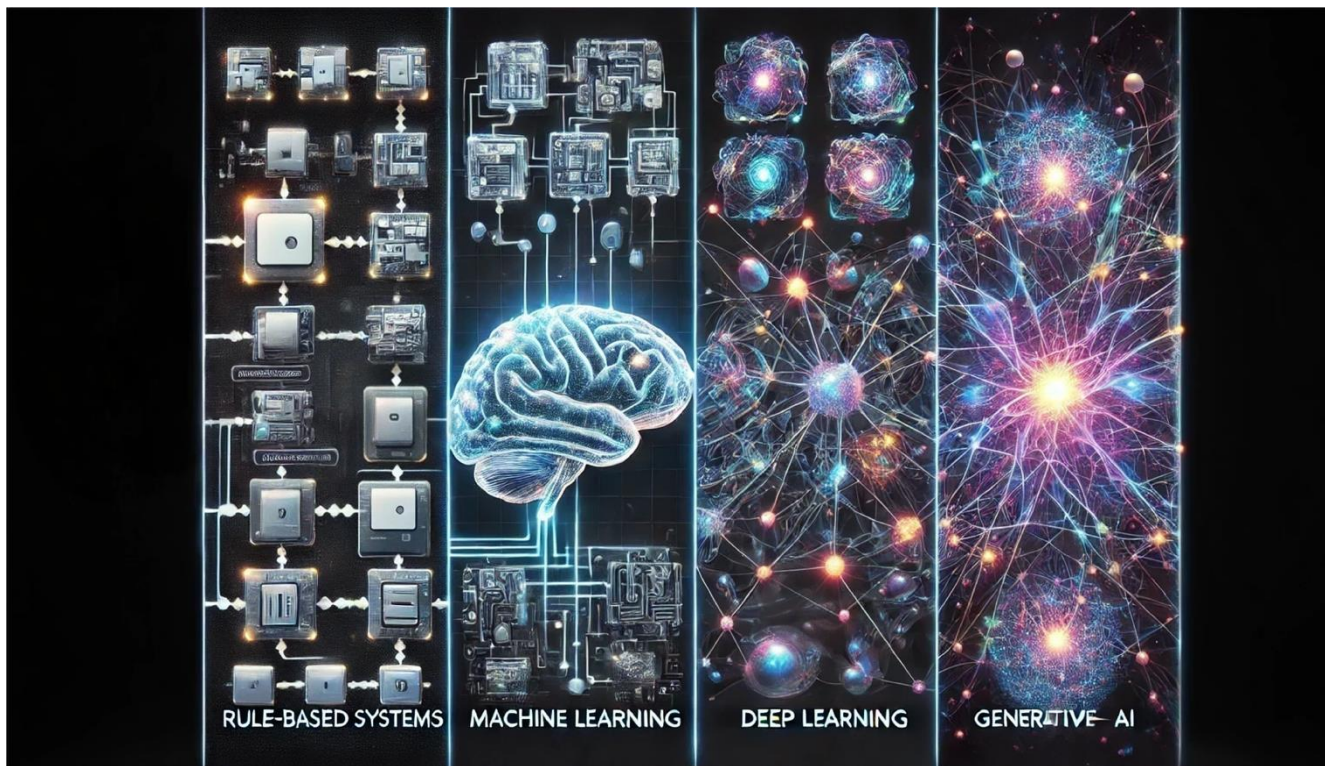
Interesting but weird - what are these people doing?



Please create new versions without human figures. Something more conceptual



Final Figure: The different stages of artificial intelligence



After ~30
prompts

Figure 1. The different stages of artificial intelligence. This image was created (with significant prompt iteration and targeted edit requests) by DALL-E, an AI image generator.



Looking Forward to AI and Medicine: Where Are We, and Where Are We Going?

by Adam Wilcox, PhD, Malachi Griffith, PhD & Obi Griffith, PhD



We speculate that the integration of AI, particularly generative AI, into medicine is expected to progress at a slower pace than in fields such as research.

Abstract

Artificial intelligence (AI) has emerged as a significant area of interest in medicine, with the potential to influence various aspects of health care. However, the real benefits of applications of AI to medicine often become obscured by the considerable attention, and hype, around its capabilities. To better understand AI's role in medicine, it is important to contextualize its development and review the stages of AI evolution that have contributed to its present state.

Introduction

My (AW) first exposure to AI was taking a course in the topic as part of my graduate studies in a new field called "Medical Informatics." Most of the students in the class were training as computer scientists, and during that semester were particularly interested in a chess match between then-world chess champion Garry Kasparov and IBM's Deep Blue supercomputer. Though Kasparov won that match, the next year Deep Blue would be the victor, marking the first time a computer program defeated a world champion in a tournament-style match. What I didn't recognize at the time was that I wasn't witnessing the beginning of a new era of AI capability, but rather the apex achievement of a rule-based era in the field that was rapidly declining. By the

time I completed research for my dissertation, both I and much of the world had moved on to a new approach to AI known as machine learning.¹ These transition points of different eras in the field are important for understanding both where it has been and the significance of where it may be going.

Artificial Intelligence History

The earliest applications of AI that demonstrated practical capabilities beyond proof-of-concepts were rule-based systems. Initially, such systems used simple rules and logic to simulate human decision-making. By the 1970s, expert systems were developed that used large sets of rules to make decisions in specialized domains. Perhaps the peak expert system was Deep Blue. With a complex but well-defined problem and sufficient computing power to search and apply rules it could outperform any human in that task. However, in general these systems were limited in their ability to handle ambiguity, learn from data, and adapt to complex problems. Their dependence on hand-crafted rules made these systems inflexible and difficult to scale.

Years later, after considerable disappointment in rule-based systems, and disinterest in AI generally, another stage of AI emerged in the 1990s with machine learning. Machine learning allowed systems to learn from data, so that

rules could be created and managed flexibly according to the data that were already collected and expected to be available in their application. To a degree, machine learning is a form of automated analysis of data, where algorithms that mimic different approaches to analyzing data are applied automatically. This made them much more flexible and scalable, though they were dependent on both the availability of training data and processing capabilities of the computers generating and applying the developed models. If sufficient data were not available to indicate patterns for learning, machine learning algorithms were limited in what could be discovered.²

The next major breakthrough came in the 2010s with the rise of deep learning, which is a subset of machine learning.³ This was in large part due to the rapid growth of available data that could be used for machine learning and increased processing capabilities; deep learning coincided with the era of "Big Data" and the emergence of graphics processing units (GPUs) for processing of data. Deep learning used a particular type of machine learning algorithms, artificial neural networks that were composed of multiple layers. These networks were capable of learning high-level features from raw data, which allowed greater abstraction and more powerful modeling of complex datasets.

In addition, neural network models once developed are highly efficient, because they use direct numeric calculations with the variables among the nodes in the networks. With more data and processing capabilities, deep learning allowed discovery of patterns that could not be identified during the previous machine learning era without substantial use of domain knowledge in the process.⁴ Not surprisingly, the areas of greatest impact with deep learning were in the domains with the largest amounts of available data, including image and speech recognition, natural language processing, and autonomous systems like self-driving cars. These data sources were available due to growth of image capture, increase of text information ability, and improved data collection.

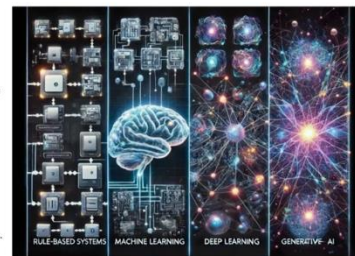


Figure 1. The different stages of artificial intelligence. This image was created (with significant prompt iteration and targeted edit requests) by DALL-E, an AI image generator.

The current era of AI, coming to prominence in the 2020s, is defined by generative AI, which extends the capabilities of AI models beyond pattern recognition and prediction to the creation of new data (Figure 1). Generative AI models such as large language models (e.g., ChatGPT) are built on transformer architectures, which are a type of deep learning model that is more efficient and powerful than previous models.⁵ This allows higher processing capability, making generative AI algorithms the most advanced that have been applied. Importantly, generative AI provides a different kind of interaction than previous modes of AI like deep learning, which were applied to specialized domains and workflows. Because generative AI creates new content rather than simply classifying content already defined, it can provide more immediate value and assist in tasks that are directly useful to a wide range of users. In fact, almost immediately after its release in late 2023, ChatGPT became the most widely and rapidly adopted technology ever.⁶

Artificial Intelligence in Medicine

The application of AI to medicine has evolved alongside the broader field of AI, though progress has often been shaped by the availability of data, regulatory constraints, and the complexity of medical tasks. Expert systems were widely developed for medical settings because of their potential to standardize complex



Agentic Coding Environments (Warp, Claude Code) moves development from typing code to coding by prompt

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The Agentic Development Environment

Warp is the fastest way to build with multiple AI agents—from writing code to shipping it. The best overall coding and terminal agent.

[Download for Mac](#)

```
$ brew install --cask warp
```

```
~/Projects/warp-server  
I want to fix the null pointer exception in this  
Sentry issue: https://warpsdev.sentry.io/  
issues/8642561740. don't run the tests yet  
I'll help you fix the null pointer exception in  
that Sentry issue. First, let me get the details  
of the issue and then analyze the code to  
identify and fix the problem.  
✓ MCP Tool: get_issue_details ({'issueUrl': '>  
Based on the Sentry issue details, I can see  
this is a null pointer exception in the  
GiveUpNotebookEditAccess GraphQL mutation. The  
error occurs in the GraphQLHandlerV2.func panic  
handler, and specifically in the  
(*sqlTx).Rollback operation, which suggests the  
database transaction object is nil.  
Let me examine the relevant source files to  
understand the issue better:  
✓ /Users/ben@ben-mac/Projects/warp-server/router/handlers/  
graphql_v2.go (140-150)  
Thinking ...
```

```
code is open in Warp  
In code server  
edit code.svg app/assets/images/logo.svg  
~/Project: code_pane.rs app/assets/images/logo.svg  
~/Project: code_02.svg app/assets/images/logo.svg  
~/Project: code_01.svg app/assets/images/logo.svg  
add an eval to {code: e}
```



Create a docker environment for hands-on bioinformatics workshop

- We host a 3 - 5 day workshop on RNAseq/scRNAseq analysis
- AWS AMI allows students to work through exercises in a consistent/robust environment
 - Requires AWS account (\$\$\$) and prevents self-organized learning and later review/practice
- A docker image with all dependencies would make more broadly accessible year round
 - Dozens of tools, linux dependencies, web server, etc



AWS Setup

[« Integrated Assignment Answers](#)

[Course](#)

[GCP Setup »](#)

Amazon AWS/AMI setup for use in workshop

This tutorial explains how Amazon cloud instances were configured for the course. This exercise is not to be completed by the students but is provided as a reference for future course developers that wish to conduct their hands on exercises on Amazon AWS.

Create AWS account

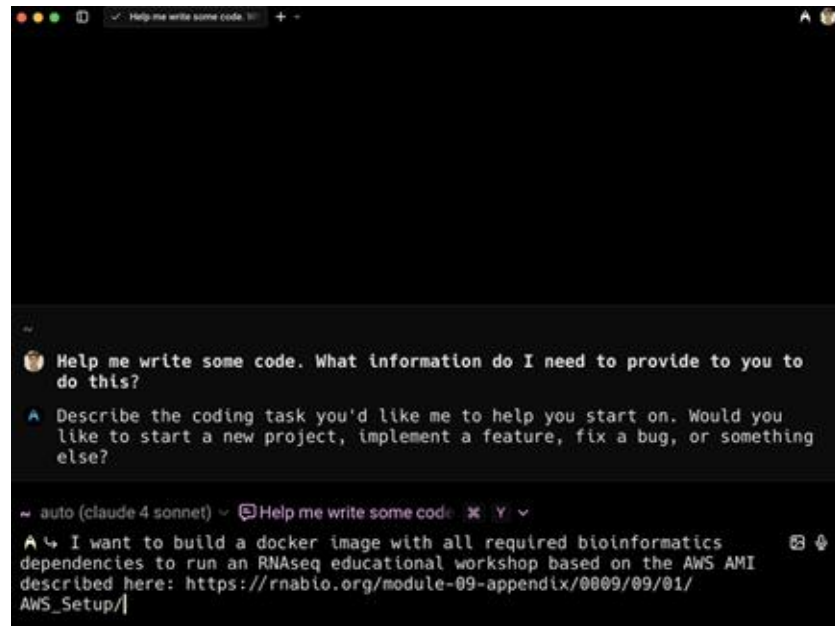
A helpful tutorial can be found [here](#)

https://rnabio.org/module-09-appendix/0009/09/01/AWS_Setup/



Create a docker environment for hands-on bioinformatics workshop

- Point warp to an existing github repo and (very prelim) attempt at a dockerfile
- Explain that build time and image size should be optimized
- Warp starts building!
 - Review each change carefully
 - Make corrections advice as needed
 - Remind to commit changes along the way
 - Ask for versioning
 - ...



Create a docker environment for hands-on bioinformatics workshop

[rnabio.org](#) / [docker](#) / [course](#) / [dockerfile_approach](#) /



obigriffith update version examples in readme ✓

fcb3c33 · 2 weeks ago History

Name	Last commit message	Last commit date
..		
CHANGELOG.md	Update CHANGELOG and VERSION	2 weeks ago
Dockerfile	Add ggridges R package and enable man pages for educational use	2 weeks ago
README.md	update version examples in readme	2 weeks ago
VERSION	Update CHANGELOG and VERSION	2 weeks ago
build.sh	Add comprehensive versioning system for Docker image	3 weeks ago
release.sh	Update release script with better GitHub release template for monorepo	3 weeks ago
version.sh	improve release instructions	2 weeks ago

README.md

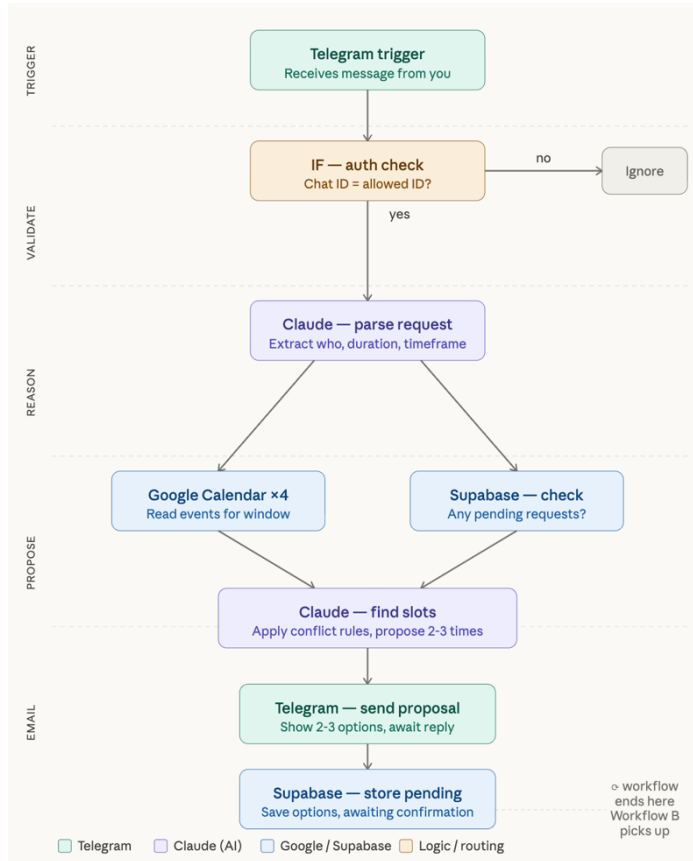


RNA-seq Bioinformatics Toolkit

This Docker image contains a comprehensive set of bioinformatics tools for RNA-seq analysis, based on the requirements described in the AWS Setup guide from [rnabio.org](#). Currently tools are limited to those needed for bulk RNAseq parts of the course.



Build your own custom agentic workflows



Caveat emptor - you still need your human brain (for now)

- I have a whole other lecture on cautionary tales and concerns
 - Reliability - Hallucinations, sycophancy, digital gaslighting
 - Ethical concerns - environmental, intellectual property, security, reproducibility, privacy
 - Education - developing critical thinking skills is still the hard work to be done



The role of AI in genomic medicine

- AI is likely to play an increasing role in nearly every aspect of genome-guided (and other types of) medicine
- AI has great potential to increase predictive accuracy, improve reproducibility, automate laborious tasks
- General AI is not yet attained (attainable?) - narrow AI applications will mostly augment/assist physicians and scientists in their work



Thank you!

